

Z Specification of Fragmenta

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November 14, 2025

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1 Generics

section *Fragmenta_Generics* **parents** *standard_toolkit*

| $\mathbb{R} : \mathbb{P}\mathbb{A}$

$\text{acyclic}[X] == \{r : X \leftrightarrow X \mid r^+ \cap \text{id } X = \emptyset\}$

$\text{connected}[X] == \{r : X \leftrightarrow X \mid \forall x : \text{dom } r; y : \text{ran } r \bullet x \mapsto y \in r^+\}$

$\text{tree}[X] == \{r : X \leftrightarrow X \mid r \in \text{acyclic} \wedge r \in X \rightarrow X\}$

$\text{forest}[X] == \{r : X \leftrightarrow X \mid r \in \text{acyclic} \wedge (\forall s : X \leftrightarrow X \mid s \subseteq r \wedge s \in \text{connected} \bullet s \in \text{tree})\}$

$\text{injrel}[X, Y] == \{r : X \leftrightarrow Y \mid r^\sim \in Y \rightarrow X\}$

$\text{antireflexive}[X] == \{r : X \leftrightarrow X \mid r \cap (\text{id } X) = \emptyset\}$

$[X, Y, Z]$ $\text{flip} : (X \rightarrow Y \rightarrow Z) \rightarrow (Y \rightarrow X \rightarrow Z)$ $\forall f : X \rightarrow Y \rightarrow Z \bullet \text{flip } f = (\lambda y : Y \bullet \lambda x : X \bullet f x y)$
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$[X, Y, Z, W]$ $\text{apply} : (X \rightarrow Z) \rightarrow (Y \rightarrow W) \rightarrow (X \times Y) \rightarrow (Z \times W)$ $\forall f : X \rightarrow Z; g : Y \rightarrow W; x : X; y : Y \bullet \text{apply } f g (x, y) = (f x, g y)$
--

relation($- =_p -$)
relation($- \subseteq_p -$)

$[X, Y]$ $- =_p - : \mathbb{P}((\mathbb{P} X \times \mathbb{P} Y) \times (\mathbb{P} X \times \mathbb{P} Y))$ $- \subseteq_p - : \mathbb{P}((\mathbb{P} X \times \mathbb{P} Y) \times (\mathbb{P} X \times \mathbb{P} Y))$ $\forall xs, zs : \mathbb{P} X; ys, ws : \mathbb{P} Y \bullet (xs, ys) =_p (zs, ws) \Leftrightarrow xs = zs \wedge ys = ws$ $\forall xs, zs : \mathbb{P} X; ys, ws : \mathbb{P} Y \bullet (xs, ys) \subseteq_p (zs, ws) \Leftrightarrow xs \subseteq zs \wedge ys \subseteq ws$
--

$[X, Y]$
$\text{map} : (X \rightarrow Y) \rightarrow \mathbb{P} X \rightarrow \mathbb{P} Y$ $\text{mapS} : (X \rightarrow Y) \rightarrow \text{seq } X \rightarrow \text{seq } Y$
$\forall f : X \rightarrow Y \bullet \text{map } f \{\} = \{\}$ $\forall f : X \rightarrow Y; x : X; xs : \mathbb{P} X \bullet \text{map } f (\{x\} \cup xs) = \{f x\} \cup (\text{map } f xs)$ $\forall f : X \rightarrow Y \bullet \text{mapS } f \langle \rangle = \langle \rangle$ $\forall f : X \rightarrow Y; x : X; xs : \text{seq } X \bullet \text{mapS } f (\langle x \rangle \frown xs) = \langle f x \rangle \frown (\text{mapS } f xs)$

function 10 **leftassoc** $(_ \boxtimes _)$

$[X]$
$_ \boxtimes _ : ((X \rightarrow X) \times \mathbb{P} X) \rightarrow (X \rightarrow X)$
$\forall f : X \rightarrow X; s : \mathbb{P} X \bullet f \boxtimes s = (\text{id } s) \oplus f$

function $(_^\oplus)$

$[X]$
$_^\oplus : (X \leftrightarrow X) \rightarrow (X \leftrightarrow X)$
$\forall r : X \leftrightarrow X \bullet r^\oplus = \text{if } r \oplus r \circ r = r \text{ then } r \text{ else } (r \oplus r \circ r)^\oplus$

$\text{opt}[X] == \{s : \mathbb{P} X \mid \# s \leq 1\}$

$[X]$
$\text{the} : \text{opt}[X] \rightarrow X$
$\forall x : X \bullet \text{the } \{x\} = x$

$[X, Y]$
$\text{flatten} : (X \rightarrow \mathbb{P} Y) \rightarrow (X \leftrightarrow Y)$
$\forall f : X \rightarrow \mathbb{P} Y \bullet \text{flatten } f = \{x : \text{dom } f; y : Y \mid y \in f x\}$

2 Graphs

section *Fragmenta_Graphs* **parents** *standard_toolkit, Fragmenta_Generics*

$[V, E]$

$Gr == \{vs : \mathbb{P} V; es : \mathbb{P} E; s, t : E \rightarrowtail V \mid s \in es \rightarrow vs \wedge t \in es \rightarrow vs\}$

$Ns : Gr \rightarrow \mathbb{P} V$ $Es : Gr \rightarrow \mathbb{P} E$ $src, tgt : Gr \rightarrow E \rightarrowtail V$	$\forall vs : \mathbb{P} V; es : \mathbb{P} E; s : E \rightarrowtail V; t : E \rightarrowtail V \bullet Ns(vs, es, s, t) = vs$ $\forall vs : \mathbb{P} V; es : \mathbb{P} E; s : E \rightarrowtail V; t : E \rightarrowtail V \bullet Es(vs, es, s, t) = es$ $\forall vs : \mathbb{P} V; es : \mathbb{P} E; s : E \rightarrowtail V; t : E \rightarrowtail V \bullet src(vs, es, s, t) = s$ $\forall vs : \mathbb{P} V; es : \mathbb{P} E; s : E \rightarrowtail V; t : E \rightarrowtail V \bullet tgt(vs, es, s, t) = t$
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$nNatS, nIntS, nRealS : V$ $to\mathbb{Z} : V \rightarrow \text{opt}[\mathbb{Z}]$ $to\mathbb{R} : V \rightarrow \text{opt}[\mathbb{R}]$	
--	--

$\emptyset_G : Gr$	$\emptyset_G = (\emptyset, \emptyset, \emptyset, \emptyset)$
--------------------	--

$Els : Gr \rightarrow (\mathbb{P} V \times \mathbb{P} E)$ $EsId : Gr \rightarrow \mathbb{P} E$	$\forall G : Gr \bullet Els\ G = (Ns\ G, Es\ G)$ $\forall G : Gr \bullet EsId\ G = \{e : Es\ G \mid src\ G\ e = tgt\ G\ e\}$
---	---

relation(*adjacent* $_$)

$adjacent_ : \mathbb{P}(Gr \times V \times V)$	$\forall G : Gr; v_1, v_2 : V \bullet adjacent(G, v_1, v_2) \Leftrightarrow \exists e : Es\ G \bullet src\ G\ e = v_1 \wedge tgt\ G\ e = v_2$
---	---

relation($adjacent_E _$)

$$\frac{adjacent_E _ : \mathbb{P}(Gr \times E \times E)}{\forall e_1, e_2 : E; \ G : Gr \bullet adjacent_E(G, e_1, e_2) \Leftrightarrow tgt\ G\ e_1 = src\ G\ e_2}$$

function 10 **leftassoc** ($_ \circ \multimap_{Es} _$)

$$\frac{_ \circ \multimap_{Es} _ : Gr \times \mathbb{P}\ V \rightarrow \mathbb{P}\ E}{\forall G : Gr; \ vs : \mathbb{P}\ V \bullet G \circ \multimap_{Es} vs = (src\ G) \sim \langle vs \rangle \cup (tgt\ G) \sim \langle vs \rangle}$$

function 10 **leftassoc** ($_ \bullet \leftrightarrow _$)

$$\frac{_ \bullet \leftrightarrow _ : Gr \times \mathbb{P}\ V \rightarrow \mathbb{P}\ E}{\forall G : Gr; \ vs : \mathbb{P}\ V \bullet G \bullet \leftrightarrow vs = (src\ G) \sim \langle vs \rangle \cap (tgt\ G) \sim \langle vs \rangle}$$

function 10 **leftassoc** ($_ \circ \multimap_{Ns} _$)

$$\frac{_ \circ \multimap_{Ns} _ : Gr \times \mathbb{P}\ E \rightarrow \mathbb{P}\ V}{\forall G : Gr; \ es : \mathbb{P}\ E \bullet G \circ \multimap_{Ns} es = \text{ran}(es \triangleleft src\ G) \cup \text{ran}(es \triangleleft tgt\ G)}$$

function 10 **leftassoc** ($_ \bowtie_{Es} _$)

$$\frac{_ \bowtie_{Es} _ : Gr \times \mathbb{P}\ E \rightarrow Gr}{\forall G : Gr; \ es : \mathbb{P}\ E \bullet G \bowtie_{Es} es = (G \circ \multimap_{Ns} es, Es\ G \cap es, es \triangleleft src\ G, es \triangleleft tgt\ G)}$$

function 10 **leftassoc** ($_ \bowtie_{Ns} _$)

$$\begin{array}{|l}
\hline
- \bowtie_{Ns} - : Gr \times \mathbb{P} V \rightarrow Gr \\
\hline
\forall G : Gr; \quad vs : \mathbb{P} V \bullet \\
\quad G \bowtie_{Ns} vs = (Ns \ G \cap vs, G \bullet \leftrightarrow \bullet \ vs, (G \bullet \leftrightarrow \bullet \ vs) \triangleleft src \ G, (G \bullet \leftrightarrow \bullet \ vs) \triangleleft tgt \ G)
\end{array}$$

function 10 **leftassoc** $(- \ominus_{Ns} -)$

$$\begin{array}{|l}
\hline
- \ominus_{Ns} - : Gr \times \mathbb{P} V \rightarrow Gr \\
\hline
\forall G : Gr; \quad vs : \mathbb{P} V \bullet \\
\quad G \ominus_{Ns} vs = (Ns \ G \setminus vs, Es \ G \setminus (G \circ \multimap_{Es} \ vs), (G \circ \multimap_{Es} \ vs) \triangleleft src \ G, (G \circ \multimap_{Es} \ vs) \triangleleft tgt \ G)
\end{array}$$

$$\begin{array}{|l}
\hline
successors : V \times Gr \rightarrow \mathbb{P} V \\
\hline
\forall v : V; \quad G : Gr \bullet \quad successors(v, G) = \{v_1 : Ns \ G \mid adjacent(G, v, v_1)\}
\end{array}$$

function $(- \varpi)$

$$\begin{array}{|l}
\hline
- \varpi : Gr \rightarrow Gr \\
\hline
\forall G : Gr \bullet \quad G \varpi = (Ns \ G, Es \ G, tgt \ G, src \ G)
\end{array}$$

function $(- \leftrightarrow)$

$$\begin{array}{|l}
\hline
- \leftrightarrow : Gr \rightarrow V \leftrightarrow V \\
\hline
\forall G : Gr \bullet \quad G \leftrightarrow = \{v_1, v_2 : Ns \ G \mid adjacent(G, v_1, v_2)\}
\end{array}$$

function $(- \leftrightarrow_E)$

$$\begin{array}{|l}
\hline
- \leftrightarrow_E : Gr \rightarrow E \leftrightarrow E \\
\hline
\forall G : Gr \bullet \quad G \leftrightarrow_E = \{e_1, e_2 : Es \ G \mid adjacent_E(G, e_1, e_2)\}
\end{array}$$

relation $(\otimes -)$

$$\frac{\otimes_- : \mathbb{P} \ Gr}{\forall G : Gr \bullet \otimes G \Leftrightarrow G \stackrel{\leftrightarrow E}{\in} \text{acyclic}}$$

relation($\boxminus_{Es} -$)
relation($\boxminus -$)

$$\frac{\boxminus_{Es} -, \boxminus_- : \mathbb{P}(Gr \times Gr)}{\begin{array}{l} \forall G_1, G_2 : Gr \bullet \boxminus_{Es}(G_1, G_2) \Leftrightarrow Es\ G_1 \cap Es\ G_2 = \emptyset \\ \forall G_1, G_2 : Gr \bullet \boxminus(G_1, G_2) \Leftrightarrow Ns\ G_1 \cap Ns\ G_2 = \emptyset \wedge \boxminus_{Es}(G_1, G_2) \end{array}}$$

relation($\boxplus -$)

$$\frac{[I]}{\frac{\boxplus_- : \mathbb{P}(I \rightarrow Gr)}{\forall Gs : I \rightarrow Gr \bullet \boxplus Gs \Leftrightarrow \forall i, j : \text{dom } Gs \mid i \neq j \bullet \boxminus(Gs\ i, Gs\ j)}}$$

function 10 **leftassoc** ($- \cup_G -$)

$$\frac{- \cup_G - : Gr \times Gr \rightarrow Gr}{\forall G_1, G_2 : Gr \bullet G_1 \cup_G G_2 = (Ns\ G_1 \cup Ns\ G_2, Es\ G_1 \cup Es\ G_2, src\ G_1 \cup src\ G_2, tgt\ G_1 \cup tgt\ G_2)}$$

function 10 **leftassoc** ($- \odot -$)

$$\frac{- \odot - : Gr \times (V \leftrightarrow V) \rightarrow Gr}{\begin{array}{l} \forall G : Gr; s : V \leftrightarrow V \mid s \in Ns\ G \rightarrow Ns\ G \wedge s \in \text{antireflexive} \bullet \\ G \odot s = (Ns\ G \setminus \text{dom } s, Es\ G, (s \boxtimes Ns\ G) \circ (src\ G), (s \boxtimes Ns\ G) \circ (tgt\ G)) \end{array}}$$

$$GrM == (V \rightarrow V) \times (E \rightarrow E)$$

$$\begin{array}{|l}
fV : GrM \rightarrow V \rightarrow V \\
fE : GrM \rightarrow E \rightarrow E \\
\hline
\forall fv : V \rightarrow V; fe : E \rightarrow E \bullet fV(fv, fe) = fv \\
\forall fv : V \rightarrow V; fe : E \rightarrow E \bullet fE(fv, fe) = fe
\end{array}$$

$$\begin{array}{|l}
gid : Gr \rightarrow GrM \\
\hline
\forall G : Gr \bullet gid\ G = (id\ (Ns\ G), id\ (Es\ G))
\end{array}$$

$$\begin{array}{|l}
\emptyset_{GM} : GrM \\
\hline
\emptyset_{GM} = (\{\}, \{\})
\end{array}$$

$$\begin{array}{|l}
domg, codg : GrM \rightarrow (\mathbb{P}\ V \times \mathbb{P}\ E) \\
\hline
\forall m : GrM \bullet domg\ m = (dom(fV\ m), dom(fE\ m)) \\
\forall m : GrM \bullet codg\ m = (ran(fV\ m), ran(fE\ m))
\end{array}$$

function 10 leftassoc $(- \cup_{GM} -)$

$$\begin{array}{|l}
- \cup_{GM} - : GrM \times GrM \rightarrow GrM \\
\bigcup_{GM} : \mathbb{P}\ GrM \rightarrow GrM \\
\hline
\forall f, g : GrM \bullet f \cup_{GM} g = (fV\ f \cup fV\ g, fE\ f \cup fE\ g) \\
\bigcup_{GM} \emptyset = \emptyset_{GM} \\
\forall f : GrM; fs : \mathbb{P}\ GrM \bullet \bigcup_{GM} (\{f\} \cup fs) = f \cup_{GM} (\bigcup_{GM} fs)
\end{array}$$

function 10 leftassoc $(- \rightarrow_G -)$

$$\begin{array}{|l}
- \rightarrow_G - : Gr \times Gr \rightarrow \mathbb{P}\ GrM \\
\hline
\forall G_1, G_2 : Gr \bullet G_1 \rightarrow_G G_2 = \{fv : Ns\ G_1 \rightarrow Ns\ G_2; fe : Es\ G_1 \rightarrow Es\ G_2 \mid \\
src\ G_2 \circ fe = fv \circ src\ G_1 \wedge tgt\ G_2 \circ fe = fv \circ tgt\ G_1\}
\end{array}$$

function 10 leftassoc $(- \rightarrow_G -)$

$$\frac{- \rightarrow_G - : Gr \times Gr \rightarrow \mathbb{P} GrM}{\forall G_1, G_2 : Gr \bullet G_1 \rightarrow_G G_2 = \{fv : Ns G_1 \rightarrow Ns G_2; fe : Es G_1 \rightarrow Es G_2 \mid \\ src G_2 \circ fe = fv \circ ((\text{dom } fe) \triangleleft (src G_1)) \wedge tgt G_2 \circ fe = fv \circ ((\text{dom } fe) \triangleleft (tgt G_1))\}}$$

function 10 **leftassoc** $(- \circ_G -)$

$$\frac{- \circ_G - : GrM \times GrM \rightarrow GrM}{\forall g, f : GrM \bullet g \circ_G f = (fV g \circ fV f, fE g \circ fE f)}$$

3 Graphs with typing

section *Fragmenta_GrswT* **parents** *standard_toolkit, Fragmenta_Generics, Fragmenta_Graphs*

$$GrwT == \{G : Gr; t : GrM \mid \text{dom } g t =_p \text{Els } G\}$$

function $(-^G)$
function $(-^T)$

$$\frac{\begin{array}{l} -^G : GrwT \rightarrow Gr \\ -^T : GrwT \rightarrow GrM \end{array}}{\begin{array}{l} \forall G : Gr; t : GrM \bullet (G, t)^G = G \\ \forall G : Gr; t : GrM \bullet (G, t)^T = t \end{array}}$$

$$\frac{\emptyset_{GrwT} : GrwT}{\emptyset_{GrwT} = (\emptyset_G, \emptyset_{GM})}$$

function 10 **leftassoc** $(- \cup_{GrwT} -)$

$$\frac{- \cup_{GrwT} - : GrwT \times GrwT \rightarrow GrwT}{\forall G_1, G_2 : GrwT \bullet G_1 \cup_{GrwT} G_2 = (G_1^G \cup_G G_2^G, G_1^T \cup_{GM} G_2^T)}$$

4 SG Element Types

section *Fragmenta_SGElemTys* **parents** *standard_toolkit, Fragmenta_Generics*

$SGNT ::= nnrml \mid nabst \mid nprxy \mid nenum \mid nval \mid nvirt$
 $SGED ::= dbi \mid duni$
 $SGET ::= einh \mid ecomp \langle\langle SGED \rangle\rangle \mid erel \langle\langle SGED \rangle\rangle \mid eder \mid epath \mid evcnt$

relation($- \prec_{NT} -$)

$\prec_{NT}$, an ordering relation on $SGNT$, indicates the node types that are inheritance-relatable:

$$\frac{}{- \prec_{NT} - : SGNT \leftrightarrow SGNT} \quad \frac{}{\forall nt_1, nt_2 : SGNT \bullet nt_1 \prec_{NT} nt_2 \Leftrightarrow nt_1 \neq nprxy \wedge nt_2 \neq nval \wedge nt_1 = nabst \Rightarrow nt_2 \in \{nabst, nvirt, nprxy\} \wedge (nt_1 = nenum \Rightarrow nt_2 \in \{nvirt, nprxy\})}$$

Above, $\prec_{NT}$ stipulates that (i) proxies must not inherit, (ii) values must not be inherited, (iii) abstract nodes may inherit from abstract and virtuals only, (iv) enumerations may only inherit virtuals or proxies

$\leq_{rNT}$ and $\leq_{ET}$ are ordering relations on sets $SGNT$ and $SGET$, respectively, indicating node and edge types that can be refinement-related, respectively:

relation($- \leq_{rNT} -$)

$$\frac{}{- \leq_{rNT} - : SGNT \leftrightarrow SGNT} \quad \frac{}{\forall nt_1, nt_2 : SGNT \bullet nt_1 \leq_{rNT} nt_2 \Leftrightarrow nt_1 = nt_2 \vee nt_1 = nprxy \vee nt_1 = nnrml \wedge nt_2 = nprxy \vee nt_2 \in \{nabst, nvirt\} \wedge nt_1 \in \{nnrml, nvirt, nabst\} \vee nt_2 = nnrml}$$

relation($- =_{ET} -$)

$$\frac{}{- =_{ET} - : SGET \leftrightarrow SGET} \quad \frac{}{\forall et_1, et_2 : SGET \bullet et_1 =_{ET} et_2 \Leftrightarrow et_1 = et_2 \vee (\forall d_1, d_2 : SGED \bullet et_1 = erel d_1 \wedge et_2 = erel d_2 \vee et_1 = ecomp d_1 \wedge et_2 = ecomp d_2)}$$

relation($- \leq_{ET} -$)

$$\frac{}{- \leq_{ET} - : SGET \leftrightarrow SGET} \quad \frac{}{\forall et_1, et_2 : SGET \bullet et_1 \leq_{ET} et_2 \Leftrightarrow \neg \{einh, eder, epath, evcnt\} \subseteq \{et_1, et_2\} \wedge (et_1 =_{ET} et_2 \vee et_1 \in \text{dom}(ecomp \sim) \wedge et_2 = erel dbi \vee et_1 = ecomp duni \wedge et_2 = erel duni)}$$

5 Multiplicities

section *Fragmenta_Mult* **parents** *standard_toolkit*, *Fragmenta_Generics*,
Fragmenta_SGElemTys

$MultVal ::= \mathbf{v} \langle \mathbb{N} \rangle \mid *$

$MultC ::= mr \langle \mathbb{N} \times MultVal \rangle \mid ms \langle MultVal \rangle$

relation $(_ =_{mv} _)$

$_ =_{mv} _ : MultVal \leftrightarrow MultVal$	$\forall m_1, m_2 : MultVal \bullet m_1 =_{mv} m_2 \Leftrightarrow \{m_1, m_2\} \subseteq \{*\} \vee \exists n : \mathbb{N} \bullet m_1 = \mathbf{v} \ n \wedge m_2 = \mathbf{v} \ n$
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function 10 **leftassoc** $(_ *_{mv} _)$

$_ *_{mv} _ : MultVal \times MultVal \rightarrow MultVal$	$\forall m : MultVal \bullet * *_{mv} m = *$ $\forall m : MultVal \bullet m *_{mv} * = *$ $\forall n_1, n_2 : \mathbb{N} \bullet (\mathbf{v} \ n_1) *_{mv} (\mathbf{v} \ n_2) = \mathbf{v} (n_1 * n_2)$
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$mlbn : MultC \rightarrow \mathbb{N}$ $mlb, mub : MultC \rightarrow MultVal$	$mlbn(ms \ *) = 0$ $\forall k : \mathbb{N} \bullet mlbn(ms(\mathbf{v} \ k)) = k$ $\forall k, m : \mathbb{N} \bullet mlbn(mr(k, \mathbf{v} \ m)) = k$ $\forall mc : MultC \bullet mlb \ mc = \mathbf{v} (mlbn \ mc)$ $\forall mv : MultVal \bullet mub(ms \ mv) = mv$ $\forall k, m : \mathbb{N} \bullet mlb(mr(k, \mathbf{v} \ m)) = \mathbf{v} \ m$
---	---

function 10 **leftassoc** $(_ *_{mr} _)$

$$\frac{- *_{mr} - : MultC \times MultC \rightarrow MultC}{\forall n_1, n_2 : \mathbb{N}; \quad mv_1, mv_2 : MultVal \bullet mr(n_1, mv_1) *_{mr} mr(n_2, mv_2) = mr(n_1 * n_2, mv_1 *_{mv} mv_2)}$$

relation($-\leq_{mv}-$)

$$\frac{- \leq_{mv} - : MultVal \leftrightarrow MultVal}{\begin{array}{l} \forall m_1, m_2 : MultVal \bullet m_1 =_{mv} m_2 \Leftrightarrow \{m_1, m_2\} \subseteq \{*\} \\ \quad \vee ((mlb \circ ms) m_1 = (mlb \circ ms) m_2 \wedge (mub \circ ms) m_1 = (mub \circ ms) m_2) \\ \forall m_1, m_2 : MultVal \bullet m_1 \leq_{mv} m_2 \Leftrightarrow m_2 = * \vee \exists j, k : \mathbb{N} \mid m_1 = \mathbf{v} j \wedge m_2 = \mathbf{v} k \bullet j \leq k \end{array}}$$

$$MultCOK == \{m : MultC \mid \exists lb : \mathbb{N}; \quad ub : MultVal \bullet m = mr(lb, ub) \wedge \mathbf{v} lb \leq_{mv} ub \\ \vee \exists mv : MultVal \bullet m = ms mv\}$$

$$MultCMany == \{ms *, mr(0, *)\}$$

$$Mult == \{ms : \mathbb{P}_1 MultCOK \mid \# ms > 1 \Rightarrow (\forall m : ms \bullet m \notin MultCMany)\}$$

$$\frac{\begin{array}{l} mk : \mathbb{N} \rightarrow MultC \\ mks : \mathbb{N} \rightarrow Mult \\ mrs : \mathbb{N} \times MultVal \rightarrow Mult \\ mopt : MultC \\ mopts, mmanys : Mult \end{array}}{\begin{array}{l} \forall k : \mathbb{N} \bullet mk k = ms(\mathbf{v} k) \wedge mks k = \{mk k\} \\ mopt = mr(0, \mathbf{v} 1) \wedge mopts = \{mopt\} \wedge mmanys = \{ms *\} \\ \forall lb : \mathbb{N}; \quad ub : MultVal \bullet mrs(lb, ub) = \{mr(lb, ub)\} \end{array}}$$

$$MultMany == \{s : Mult \mid \exists m : MultC \bullet s = \{m\} \wedge m \in MultCMany\}$$

$$MultRange == \{s : Mult \mid \exists m : MultC \bullet s = \{m\} \wedge (\exists k : \mathbb{N} \mid k > 1 \bullet m = ms(\mathbf{v} k) \\ \vee \exists lb : \mathbb{N}; \quad umv : MultVal \mid \mathbf{v} 2 \leq_{mv} umv \bullet m = mr(lb, umv))\}$$

$$MultEither == \{s : Mult \mid \# s > 1\}$$

$$MultLBZ == \{ms : Mult \mid \exists m : ms \bullet mlbn m = 0\}$$

relation($-\checkmark-$)
relation($-\cdots-$)

$$\frac{- \check{\bowtie} - : \mathbb{P}(\mathbb{N} \times MultC)}{\forall k : \mathbb{N}; m : MultC \bullet k \check{\bowtie} m \Leftrightarrow mlb\ m \leq_{mv} \mathbf{v}\ k \wedge \mathbf{v}\ k \leq_{mv} mub\ m}$$

$$\frac{- \cdots - : \mathbb{P}(\mathbb{N} \times Mult)}{\begin{array}{l} \forall k : \mathbb{N}; m : MultC \bullet k \cdots \{m\} \Leftrightarrow k \check{\bowtie} m \\ \forall k : \mathbb{N}; m : MultC; sms : Mult \bullet k \cdots (\{m\} \cup sms) \Leftrightarrow k \check{\bowtie} m \vee k \cdots sms \end{array}}$$

relation($-\leq_{\mathcal{M}^c}-$)

$$\frac{- \leq_{\mathcal{M}^c} - : MultC \leftrightarrow MultC}{\forall m_1, m_2 : MultC \bullet m_1 \leq_{\mathcal{M}^c} m_2 \Leftrightarrow mlb\ m_2 \leq_{mv} mlb\ m_1 \wedge mub\ m_1 \leq_{mv} mub\ m_2}$$

relation($-\leq_{\mathcal{M}}-$)

$$\frac{- \leq_{\mathcal{M}} - : Mult \leftrightarrow Mult}{\forall m_1, m_2 : Mult \bullet m_1 \leq_{\mathcal{M}} m_2 \Leftrightarrow \forall mc_1 : m_1 \bullet \exists mc_2 : m_2 \bullet mc_1 \leq_{\mathcal{M}^c} mc_2}$$

relation($-\propto-$)

$$\frac{- \propto - : \mathbb{P}(SGET \times (Mult \times Mult))}{\begin{array}{l} \forall et : SGET; m_1, m_2 : Mult \bullet et \propto (m_1, m_2) \Leftrightarrow et = erel\ dbi \vee et = eder \\ \vee et = ecomp\ duni \wedge m_1 = mks\ 1 \\ \vee et = erel\ duni \wedge m_1 \in MultMany \\ \vee et = ecomp\ dbi \wedge m_1 \in \{mks\ 1, mopts\} \end{array}}$$

relation(**rbounded**_⊥)

relation(**eitherbounded**_⊥)

$$\begin{array}{l} \overline{\overline{[X, Y]}} \\ \text{rbounded}_{\perp} : \mathbb{P}((X \leftrightarrow Y) \times \mathbb{P}X \times MultC) \\ \text{eitherbounded}_{\perp} : \mathbb{P}((X \leftrightarrow Y) \times \mathbb{P}X \times Mult) \\ \hline \forall r : X \leftrightarrow Y; s : \mathbb{P}X; m : MultC \bullet \\ \quad \text{rbounded}(r, s, m) \Leftrightarrow \forall x : s \bullet \#(r \downarrow \{x\}) \check{\bowtie} m \\ \forall r : X \leftrightarrow Y; s : \mathbb{P}X; ms : Mult \bullet \\ \quad \text{eitherbounded}(r, s, ms) \Leftrightarrow \forall x : s \bullet \#(r \downarrow \{x\}) \cdots ms \end{array}$$

relation($r\mathcal{M}Ok_$)

$[X, Y]$
$r\mathcal{M}Ok_ : \mathbb{P}((X \leftrightarrow Y) \times \mathbb{P}X \times \mathbb{P}Y \times Mult \times Mult)$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y; m : Mult \bullet r\mathcal{M}Ok(r, s, t, mks\ 0, m) \Leftrightarrow s \triangleleft r = \{\}$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y; m : Mult \bullet r\mathcal{M}Ok(r, s, t, m, mks\ 0) \Leftrightarrow r \triangleright t = \{\}$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y \bullet r\mathcal{M}Ok(r, s, t, mks\ 1, mks\ 1) \Leftrightarrow r \in s \multimap t$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y \bullet r\mathcal{M}Ok(r, s, t, mopts, mks\ 1) \Leftrightarrow r \in s \multimap t$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y \bullet r\mathcal{M}Ok(r, s, t, mks\ 1, mopts) \Leftrightarrow r \sim \in t \multimap s$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y; mm : MultMany \bullet r\mathcal{M}Ok(r, s, t, mm, mks\ 1) \Leftrightarrow r \in s \rightarrow t$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y; mm : MultMany \bullet r\mathcal{M}Ok(r, s, t, mks\ 1, mm) \Leftrightarrow r \sim \in t \rightarrow s$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y; mr : MultRange \bullet$ $r\mathcal{M}Ok(r, s, t, mr, mks\ 1) \Leftrightarrow r \in s \rightarrow t \wedge \text{rbounded}(r \sim, t, \text{the } mr)$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y; mr : MultRange \bullet$ $r\mathcal{M}Ok(r, s, t, mks\ 1, mr) \Leftrightarrow r \sim \in t \rightarrow s \wedge \text{rbounded}(r, s, \text{the } mr)$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y; me : MultEither \bullet$ $r\mathcal{M}Ok(r, s, t, me, mks\ 1) \Leftrightarrow r \in s \rightarrow t \wedge \text{eitherbounded}(r \sim, t, me)$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y; me : MultEither \bullet$ $r\mathcal{M}Ok(r, s, t, mks\ 1, me) \Leftrightarrow r \sim \in t \rightarrow s \wedge \text{eitherbounded}(r, s, me)$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y \bullet r\mathcal{M}Ok(r, s, t, mopts, mopts) \Leftrightarrow r \in s \multimap t$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y; mm : MultMany \bullet r\mathcal{M}Ok(r, s, t, mm, mopts) \Leftrightarrow r \in s \multimap t$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y; mm : MultMany \bullet$ $r\mathcal{M}Ok(r, s, t, mopts, mm) \Leftrightarrow r \sim \in t \multimap s$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y; m : MultRange \bullet$ $r\mathcal{M}Ok(r, s, t, m, mopts) \Leftrightarrow r \in s \multimap t \wedge \text{rbounded}(r \sim, t, \text{the } m)$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y; m : MultRange \bullet$ $r\mathcal{M}Ok(r, s, t, mopts, m) \Leftrightarrow r \sim \in t \multimap s \wedge \text{rbounded}(r, s, \text{the } m)$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y; m : MultEither \bullet$ $r\mathcal{M}Ok(r, s, t, m, mopts) \Leftrightarrow r \in s \multimap t \wedge \text{eitherbounded}(r \sim, t, m)$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y; m : MultEither \bullet$ $r\mathcal{M}Ok(r, s, t, mopts, m) \Leftrightarrow r \sim \in t \multimap s \wedge \text{eitherbounded}(r, s, m)$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y; mm_1, mm_2 : MultMany \bullet$ $r\mathcal{M}Ok(r, s, t, mm_1, mm_2) \Leftrightarrow r \in s \leftrightarrow t$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y; mm : MultMany; mr : MultRange \bullet$ $r\mathcal{M}Ok(r, s, t, mm, mr) \Leftrightarrow r \in s \leftrightarrow t \wedge \text{rbounded}(r, s, \text{the } mr)$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y; mm : MultMany; mr : MultRange \bullet$ $r\mathcal{M}Ok(r, s, t, mr, mm) \Leftrightarrow r \in s \leftrightarrow t \wedge \text{rbounded}(r \sim, t, \text{the } mr)$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y; mm : MultMany; me : MultEither \bullet$ $r\mathcal{M}Ok(r, s, t, mm, me) \Leftrightarrow r \in s \leftrightarrow t \wedge \text{eitherbounded}(r, s, me)$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y; mm : MultMany; me : MultEither \bullet$ $r\mathcal{M}Ok(r, s, t, me, mm) \Leftrightarrow r \in s \leftrightarrow t \wedge \text{eitherbounded}(r \sim, t, me)$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y; mr_1, mr_2 : MultRange \bullet$ $r\mathcal{M}Ok(r, s, t, mr_1, mr_2) \Leftrightarrow r \in s \leftrightarrow t \wedge \text{rbounded}(r, s, \text{the } mr_2) \wedge \text{rbounded}(r \sim, t, \text{the } mr_1)$
$\forall r : X \leftrightarrow Y; s : \mathbb{P}X; t : \mathbb{P}Y; me_1, me_2 : MultEither \bullet$ $r\mathcal{M}Ok(r, s, t, me_1, me_2) \Leftrightarrow r \in s \leftrightarrow t \wedge \text{eitherbounded}(r, s, me_2) \wedge \text{eitherbounded}(r \sim, t, me_1)$

6 Path Expressions

section *Fragmenta_PEs* **parents** *standard_toolkit, Fragmenta_Generics, Fragmenta_Graphs*

$PEA ::= \text{edg} \langle\langle E \rangle\rangle \mid \text{einv} \langle\langle E \rangle\rangle$

$PEC ::= \text{eat} \langle\langle PEA \rangle\rangle \mid \text{eresd} \langle\langle V \times PEA \rangle\rangle \mid \text{eresr} \langle\langle PEA \times V \rangle\rangle$

$PE ::= \text{ec} \langle\langle PEC \rangle\rangle \mid \text{scmp} \langle\langle PEC \times PE \rangle\rangle$

$ePEA : PEA \rightarrow E$	
$\forall e : E \bullet ePEA(\text{edg } e) = e$	
$\forall e : E \bullet ePEA(\text{einv } e) = e$	
$\text{startEA}_C : PEC \rightarrow PEA$ $\text{startEA} : PE \rightarrow PEA$	
$\forall \text{pea} : PEA \bullet \text{startEA}_C(\text{eat } \text{pea}) = \text{pea}$	
$\forall v : V; \text{pea} : PEA \bullet \text{startEA}_C(\text{eresd } (v, \text{pea})) = \text{pea}$	
$\forall v : V; \text{pea} : PEA \bullet \text{startEA}_C(\text{eresr } (\text{pea}, v)) = \text{pea}$	
$\forall \text{pec} : PEC \bullet \text{startEA}(\text{ec } \text{pec}) = \text{startEA}_C \text{ pec}$	
$\forall \text{pec} : PEC; \text{pe} : PE \bullet \text{startEA}(\text{scmp } (\text{pec}, \text{pe})) = \text{startEA}_C \text{ pec}$	
$\text{endEA}_C : PEC \rightarrow PEA$ $\text{endEA} : PE \rightarrow PEA$	
$\forall \text{pea} : PEA \bullet \text{endEA}_C(\text{eat } \text{pea}) = \text{pea}$	
$\forall v : V; \text{pea} : PEA \bullet \text{endEA}_C(\text{eresd } (v, \text{pea})) = \text{pea}$	
$\forall v : V; \text{pea} : PEA \bullet \text{endEA}_C(\text{eresr } (\text{pea}, v)) = \text{pea}$	
$\forall \text{pec} : PEC \bullet \text{endEA}(\text{ec } \text{pec}) = \text{endEA}_C(\text{pec})$	
$\forall \text{pec} : PEC; \text{pe} : PE \bullet \text{endEA}(\text{scmp } (\text{pec}, \text{pe})) = \text{endEA}_C \text{ pe}$	

$srcPEA : Gr \rightarrow PEA \rightarrow V$ $srcPEC : Gr \rightarrow PEC \rightarrow V$ $srcPE : Gr \rightarrow PE \rightarrow V$
$\forall G : Gr; e : E \bullet srcPEA G (edg e) = src G e$ $\forall G : Gr; e : E \bullet srcPEA G (einv e) = tgt G e$ $\forall G : Gr; v : V; pea : PEA \bullet srcPEC G (eat pea) = srcPEA G pea$ $\forall G : Gr; v : V; pea : PEA \bullet srcPEC G (eresd (v, pea)) = srcPEA G pea$ $\forall G : Gr; v : V; pea : PEA \bullet srcPEC G (eresr (pea, v)) = srcPEA G pea$ $\forall G : Gr; pec : PEC \bullet srcPE G (ec pec) = srcPEC G pec$ $\forall G : Gr; pec : PEC; pe : PE \bullet srcPE G (scmp (pec, pe)) = srcPEC G pec$
$tgtPEA : Gr \rightarrow PEA \rightarrow V$ $tgtPEC : Gr \rightarrow PEC \rightarrow V$ $tgtPE : Gr \rightarrow PE \rightarrow V$
$\forall G : Gr; e : E \bullet tgtPEA G (edg e) = tgt G e$ $\forall G : Gr; e : E \bullet tgtPEA G (einv e) = src G e$ $\forall G : Gr; v : V; pea : PEA \bullet tgtPEC G (eat pea) = tgtPEA G pea$ $\forall G : Gr; v : V; pea : PEA \bullet tgtPEC G (eresd (v, pea)) = tgtPEA G pea$ $\forall G : Gr; v : V; pea : PEA \bullet tgtPEC G (eresr (pea, v)) = tgtPEA G pea$ $\forall G : Gr; pec : PEC \bullet tgtPE G (ec pec) = tgtPEC G pec$ $\forall G : Gr; pec : PEC; pe : PE \bullet tgtPE G (scmp (pec, pe)) = tgtPEC G pe$
$rsrcPE : PE \rightarrow E$ $rtgtPE : PE \rightarrow E$
$rsrcPE = ePEA \circ startEA$ $rtgtPE = ePEA \circ endEA$
$esPEC : PEC \rightarrow \mathbb{P} E$ $esPE : PE \rightarrow \mathbb{P} E$
$\forall pea : PEA \bullet esPEC (eat pea) = \{ePEA pea\}$ $\forall v : V; pea : PEA \bullet esPEC (eresd (v, pea)) = \{ePEA pea\}$ $\forall v : V; pea : PEA \bullet esPEC (eresr (pea, v)) = \{ePEA pea\}$ $\forall pec : PEC \bullet esPE (ec pec) = esPEC pec$ $\forall pec : PEC; pe : PE \bullet esPE (scmp (pec, pe)) = esPEC(pec) \cup esPE(pe)$

7 Structural Graphs

section *Fragmenta_SGs* **parents** *standard_toolkit, Fragmenta_Generics, Fragmenta_Graphs, Fragmenta_SGElemTys, Fragmenta_Mult, Fragmenta_PEs, Fragmenta_GrswT*

Z Type *VCI* represents the information associated with a value constraint edge: an operator such as equality (*SGVCEOP*) and an optional edge, which is either one of the edges or the node or none if the constraint edge refers to the values of the node itself.

$$SGVCEOP ::= eq \mid neq \mid leq \mid geq \mid lt \mid gt$$

$$VCI == SGVCEOP \times opt[E]$$

$$SGr_0 == \{ G : Gr; nt : V \rightarrow SGNT; et : E \rightarrow SGET; sm, tm : E \rightarrow Mult; p : E \rightarrow PE; d : E \leftrightarrow E; vci : E \rightarrow VCI \mid nt \in Ns \ G \rightarrow SGNT \wedge et \in Es \ G \rightarrow SGET \wedge d \in Es \ G \leftrightarrow Es \ G \}$$

$gr : SGr_0 \rightarrow Gr$
 $sg_Ns : SGr_0 \rightarrow \mathbb{P} V$
 $sg_Es : SGr_0 \rightarrow \mathbb{P} E$
 $sg_src, sg_tgt : SGr_0 \rightarrow E \rightarrow V$
 $nty : SGr_0 \rightarrow V \rightarrow SGNT$
 $ety : SGr_0 \rightarrow E \rightarrow SGET$
 $srcm, tgtm : SGr_0 \rightarrow E \rightarrow Mult$
 $pe : SGr_0 \rightarrow E \rightarrow PE$
 $ds : SGr_0 \rightarrow E \leftrightarrow E$
 $vcei : SGr_0 \rightarrow E \rightarrow SGVCEOP \times opt[E]$

$\forall G : Gr; nt : V \rightarrow SGNT; et : E \rightarrow SGET; sm, tm : E \rightarrow Mult; p : E \rightarrow PE; d : E \leftrightarrow E;$
 $vci : E \rightarrow VCI \bullet gr(G, nt, et, sm, tm, p, d, vci) = G$
 $sg_Ns = Ns \circ gr$
 $sg_Es = Es \circ gr$
 $sg_src = src \circ gr$
 $sg_tgt = tgt \circ gr$
 $\forall G : Gr; nt : V \rightarrow SGNT; et : E \rightarrow SGET; sm, tm : E \rightarrow Mult; p : E \rightarrow PE; d : E \leftrightarrow E;$
 $vci : E \rightarrow VCI \bullet nty(G, nt, et, sm, tm, p, d, vci) = nt$
 $\forall G : Gr; nt : V \rightarrow SGNT; et : E \rightarrow SGET; sm, tm : E \rightarrow Mult; p : E \rightarrow PE; d : E \leftrightarrow E;$
 $vci : E \rightarrow VCI \bullet ety(G, nt, et, sm, tm, p, d, vci) = et$
 $\forall G : Gr; nt : V \rightarrow SGNT; et : E \rightarrow SGET; sm, tm : E \rightarrow Mult; p : E \rightarrow PE; d : E \leftrightarrow E;$
 $vci : E \rightarrow VCI \bullet srcm(G, nt, et, sm, tm, p, d, vci) = sm$
 $\forall G : Gr; nt : V \rightarrow SGNT; et : E \rightarrow SGET; sm, tm : E \rightarrow Mult; p : E \rightarrow PE; d : E \leftrightarrow E;$
 $vci : E \rightarrow VCI \bullet tgtm(G, nt, et, sm, tm, p, d, vci) = tm$
 $\forall G : Gr; nt : V \rightarrow SGNT; et : E \rightarrow SGET; sm, tm : E \rightarrow Mult; p : E \rightarrow PE; d : E \leftrightarrow E;$
 $vci : E \rightarrow VCI \bullet pe(G, nt, et, sm, tm, p, d, vci) = p$
 $\forall G : Gr; nt : V \rightarrow SGNT; et : E \rightarrow SGET; sm, tm : E \rightarrow Mult; p : E \rightarrow PE; d : E \leftrightarrow E;$
 $vci : E \rightarrow VCI \bullet ds(G, nt, et, sm, tm, p, d, vci) = d$
 $\forall G : Gr; nt : V \rightarrow SGNT; et : E \rightarrow SGET; sm, tm : E \rightarrow Mult; p : E \rightarrow PE; d : E \leftrightarrow E;$
 $vci : E \rightarrow VCI \bullet vcei(G, nt, et, sm, tm, p, d, vci) = vci$

$\emptyset_{SG} : SGr_0$
$\emptyset_{SG} = (\emptyset_G, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset)$
$NsTy : SGr_0 \rightarrow \mathbb{P} SGNT \rightarrow \mathbb{P} V$ $EsTy : SGr_0 \rightarrow \mathbb{P} SGET \rightarrow \mathbb{P} E$
$\forall SG : SGr_0; nts : \mathbb{P} SGNT \bullet NsTy SG nts = (nty SG) \sim \langle nts \rangle$ $\forall SG : SGr_0; ets : \mathbb{P} SGET \bullet EsTy SG ets = (ety SG) \sim \langle ets \rangle$
$EsA, EsI, EsM, EsD, EsVCnt, EsPa, EsPaCnt : SGr_0 \rightarrow \mathbb{P} E$ $EsA = (\text{flip } EsTy) (\text{erel } \langle SGED \rangle \cup \text{ecomp } \langle SGED \rangle)$ $EsI = (\text{flip } EsTy) \{eih\}$ $EsD = (\text{flip } EsTy) \{eder\}$ $EsPa = (\text{flip } EsTy) \{epath\}$ $EsVCnt = (\text{flip } EsTy) \{evcnt\}$ $\forall SG : SGr_0 \bullet EsM SG = EsA SG \cup EsD SG$ $\forall SG : SGr_0 \bullet EsPaCnt SG = EsD SG \cup EsPa SG$
$NsN, NsP, NsEther, NsVi, NsVa : SGr_0 \rightarrow \mathbb{P} V$ $NsN = (\text{flip } NsTy) \{nnrml\}$ $NsP = (\text{flip } NsTy) \{nprxy\}$ $NsEther = (\text{flip } NsTy) \{nabst, nvirt, nenum\}$ $NsVi = (\text{flip } NsTy) \{nvirt\}$ $NsVa = (\text{flip } NsTy) \{nval\}$
$tpe : SGr_0 \rightarrow E \leftrightarrow PE$ $\forall SG : SGr_0 \bullet tpe SG = \{e : sg_Es SG \bullet (e, (ec \circ eat \circ edg) e)\} \oplus pe SG$
$\mathfrak{h} : SGr_0 \rightarrow Gr$ $\prec : SGr_0 \rightarrow V \leftrightarrow V$ $\forall SG : SGr_0 \bullet \mathfrak{h} SG = gr SG \bowtie_{Es} EsI SG$ $\prec = (-^{\leftrightarrow}) \circ \mathfrak{h}$

$$\begin{array}{|l}
srcma : SGr_0 \rightarrow (E \leftrightarrow Mult) \\
\hline
\forall SG : SGr_0 \bullet srcma SG = \\
\quad (srcm SG) \oplus (EsTy SG \{ecompe duni\} \times \{mks 1\}) \oplus (EsTy SG \{erel duni\} \times \{\{ms *\}\})
\end{array}$$

relation(*MetysOk* _)

$$\begin{array}{|l}
MetysOk_ : \mathbb{P} SGr_0 \\
\hline
\forall SG : SGr_0 \bullet MetysOk SG \Leftrightarrow \forall e : EsM SG \bullet (ety SG e) \propto (srcma SG e, tgtm SG e)
\end{array}$$

$$\begin{array}{|l}
\preceq : SGr_0 \rightarrow V \leftrightarrow V \\
\hline
\forall SG : SGr_0 \bullet \preceq SG = (\prec SG)^*
\end{array}$$

$$\begin{array}{|l}
rsrc, rtgt : SGr_0 \rightarrow \mathbb{P} E \rightarrow E \leftrightarrow V \\
src_M^*, src^*, tgt_M^*, tgt^* : SGr_0 \rightarrow E \leftrightarrow V \\
\hline
\forall SG : SGr_0; es : \mathbb{P} E \bullet rsrc SG es = es \triangleleft (sg_src SG) \\
\forall SG : SGr_0; es : \mathbb{P} E \bullet rtgt SG es = es \triangleleft (sg_tgt SG) \\
\forall SG : SGr_0 \bullet src_M^* SG = (rsrc SG (EsA SG)) \mathbin{\circledast} (\preceq SG) \sim \\
\forall SG : SGr_0 \bullet src^* SG = (rsrc SG (EsA SG \cup EsPaCnt SG)) \mathbin{\circledast} (\preceq SG) \sim \\
\forall SG : SGr_0 \bullet tgt_M^* SG = (rtgt SG (EsA SG)) \mathbin{\circledast} (\preceq SG) \sim \\
\forall SG : SGr_0 \bullet tgt^* SG = (rtgt SG (EsA SG \cup EsPaCnt SG)) \mathbin{\circledast} (\preceq SG) \sim
\end{array}$$

Predicate VCntEsOk says whether the value constraint edges of a SG are well-formed.

relation(VCntEsOk_)

$$\begin{array}{|l}
VCntEsOk_ : \mathbb{P} SGr_0 \\
\hline
\forall SG : SGr_0 \bullet VCntEsOk SG \Leftrightarrow vcei SG \in EsVCnt SG \rightarrow VCI \\
\quad \wedge \bigcup (map second ((ran \circ vcei) SG)) \subseteq EsA SG \\
\quad \wedge (sg_tgt SG) \mathbin{\circledast} EsVCnt SG \subseteq NsVa SG \cup NsP SG
\end{array}$$

relation(*inhOk* _)

$$\begin{array}{|l}
inhOk_ : \mathbb{P} \, SGr_0 \\
\hline
\forall SG : SGr_0 \bullet inhOk \, SG \\
\quad \Leftrightarrow (\forall v, v' : sg_Ns \, SG \mid (v, v') \in (\prec SG) \bullet nty \, SG \, v \prec_{NT} nty \, SG \, v') \wedge \odot(\upharpoonright SG)
\end{array}$$

$$\begin{aligned}
SGr == \{ & SG : SGr_0 \mid \{srcma \, SG, tgm \, SG\} \subseteq EsM \, SG \rightarrow Mult \wedge (\text{dom} \circ pe) \, SG = EsPaCnt \, SG \\
& \wedge ds \, SG \in \text{antireflexive}[EsPa \, SG] \\
& \wedge MetysOk \, SG \wedge inhOk \, SG \wedge VCntEsOk \, SG \}
\end{aligned}$$

relation(etherealAreInherited _)

$$\begin{array}{|l}
etherealAreInherited_ : \mathbb{P} \, SGr_0 \\
\hline
\forall SG : SGr_0 \bullet \text{etherealAreInherited} \, SG \Leftrightarrow NsEther \, SG \subseteq \text{ran}(\prec SG)
\end{array}$$

relation(derInhOk _)

$$\begin{array}{|l}
derInhOk_ : \mathbb{P} \, SGr_0 \\
\hline
\forall SG : SGr_0 \bullet derInhOk \, SG \Leftrightarrow \forall e : EsD \, SG \bullet \\
\quad (gr \, SG \bowtie_{Es} esPE (pe \, SG \, e))^{\leftrightarrow} \circ (\preceq SG) \subseteq (\preceq SG) \circ (gr \, SG \bowtie_{Es} esPE (pe \, SG \, e))^{\leftrightarrow}
\end{array}$$

relation(okPEA _)
relation(okPEC _)
relation(okPE _)
relation(okPEASrc _)
relation(okPEATgt _)

$$\begin{array}{|l}
okPEASrc_ : \mathbb{P}(SGr_0 \times V \times PEA) \\
okPEATgt_ : \mathbb{P}(SGr_0 \times V \times PEA) \\
\hline
\forall SG : SGr_0; v : V; e : E \bullet okPEASrc(SG, v, edg \, e) \Leftrightarrow (e, v) \in src^* \, SG \\
\forall SG : SGr_0; v : V; e : E \bullet okPEASrc(SG, v, einv \, e) \Leftrightarrow (e, v) \in tgt^* \, SG \\
\forall SG : SGr_0; v : V; e : E \bullet okPEATgt(SG, v, edg \, e) \Leftrightarrow (e, v) \in tgt^* \, SG \\
\forall SG : SGr_0; v : V; e : E \bullet okPEATgt(SG, v, einv \, e) \Leftrightarrow (e, v) \in src^* \, SG
\end{array}$$

$okPEA_ : \mathbb{P}(SGr_0 \times PEA)$
$\forall SG : SGr_0; e : E \bullet okPEA(SG, edg\ e) \Leftrightarrow e \in sg_Es\ SG$
$\forall SG : SGr_0; e : E \bullet okPEA(SG, einv\ e) \Leftrightarrow e \in sg_Es\ SG$

$okPEC_ : \mathbb{P}(SGr_0 \times PEC)$
$okPE_ : \mathbb{P}(SGr_0 \times PE)$
$\forall SG : SGr_0; v : V; pea : PEA \bullet okPEC(SG, eat\ pea) \Leftrightarrow okPEA(SG, pea)$
$\forall SG : SGr_0; v : V; pea : PEA \bullet$ $okPEC(SG, eresd(v, pea)) \Leftrightarrow okPEA(SG, pea) \wedge okPEASrc(SG, v, pea)$
$\forall SG : SGr_0; v : V; pea : PEA \bullet$ $okPEC(SG, eresr(pea, v)) \Leftrightarrow okPEA(SG, pea) \wedge okPEATgt(SG, v, pea)$
$\forall SG : SGr_0; pec : PEC; pe : PE \bullet$ $okPE(SG, scmp(pec, pe)) \Leftrightarrow okPEC(SG, pec) \wedge okPE(SG, pe)$ $\wedge tgtPEC(gr\ SG \bowtie_{Es} EsA\ SG)\ pec = srcPE(gr\ SG \bowtie_{Es} EsA\ SG)\ pe$

relation(isVCEECnt_)
relation(isVCENCnt_)

$isVCEECnt_, isVCENCnt_ : \mathbb{P}(SGr \times E)$
$\forall SG : SGr; vce : E \bullet isVCEECnt(SG, vce) \Leftrightarrow (second \circ (vcei\ SG))\ vce \neq \emptyset$
$\forall SG : SGr; vce : E \bullet isVCENCnt(SG, vce) \Leftrightarrow (second \circ (vcei\ SG))\ vce = \emptyset$

relation(commonAncestor_)
relation(EsVCntsOk_)

$commonAncestor_ : \mathbb{P}(SGr_0 \times V \times V)$
$EsVCntsOk_ : \mathbb{P}\ SGr_0$
$\forall SG : SGr_0; n_1, n_2 : V \bullet commonAncestor(SG, n_1, n_2) \Leftrightarrow \exists nt : sg_Ns\ SG \bullet (n_1, nt) \in \preceq\ SG \wedge (n_2, nt) \in \preceq\ SG$
$\forall SG : SGr_0 \bullet EsVCntsOk\ SG \Leftrightarrow$ $\forall vce : EsVCnt\ SG \bullet isVCEECnt(SG, vce) \Rightarrow$ $commonAncestor(SG, sg_tgt\ SG((the \circ second \circ (vcei\ SG))\ vce), sg_tgt\ SG\ vce)$ $\wedge isVCENCnt(SG, vce) \Rightarrow commonAncestor(SG, sg_src\ SG\ vce, sg_tgt\ SG\ vce)$

relation(EsCntsOk_)

$$\begin{array}{|l}
\hline
EsCntsOk _ : \mathbb{P} \, SGr_0 \\
\hline
\forall SG : SGr_0 \bullet \\
\quad EsCntsOk \, SG \Leftrightarrow derInhOk \, SG \wedge \forall e : EsPaCnt \, SG \bullet okPE(SG, pe \, SG \, e) \\
\quad \wedge EsVCntsOk \, SG
\end{array}$$

relation(*inhTree* $_$)

$$\begin{array}{|l}
\hline
inhTree _ : \mathbb{P} \, SGr_0 \\
\hline
\forall SG : SGr_0 \bullet inhTree \, SG \Leftrightarrow ((\lhd SG) \ominus_{Ns} (NsVi \, SG))^{*\Rightarrow} \in \text{tree}
\end{array}$$

$TSGr == \{ SG : SGr \mid \text{etherealAreInherited } SG \wedge EsCntsOk \, SG \wedge inhTree \, SG \}$

relation($\Box_{SGs} _$)

$$\begin{array}{|l}
\hline
\Box_{SGs} _ : \mathbb{P}(SGr \times SGr) \\
\hline
\forall SG_1, SG_2 : SGr \bullet \Box_{SGs}(SG_1, SG_2) \Leftrightarrow \Box(gr \, SG_1, gr \, SG_2)
\end{array}$$

function 10 **leftassoc** ($_ \cup_{SG} _$)

$$\begin{array}{|l}
\hline
_ \cup_{SG} _ : SGr \times SGr \rightarrow SGr \\
\hline
\forall SG_1, SG_2 : SGr \bullet SG_1 \cup_{SG} SG_2 = (gr \, SG_1 \cup_G gr \, SG_2, nty \, SG_1 \cup nty \, SG_2, \\
\quad ety \, SG_1 \cup ety \, SG_2, srcm \, SG_1 \cup srcm \, SG_2, tgtm \, SG_1 \cup tgtm \, SG_2, pe \, SG_1 \cup pe \, SG_2, ds \, SG_1 \cup ds \, SG_2, vcei \, SG_1 \cup vcei \, SG_2)
\end{array}$$

function 10 **leftassoc** ($_ \odot^{SG} _$)

$$\begin{array}{|l}
\hline
_ \odot^{SG} _ : SGr \times (V \rightarrow V) \rightarrow SGr \\
\hline
\forall SG : SGr; s : V \rightarrow V \mid s \in NsP \, SG \rightarrow sg_Ns \, SG \bullet \\
\quad SG \odot^{SG} s = (gr \, SG \odot s, (\text{dom } s \setminus \text{ran } s) \triangleleft nty \, SG, ety \, SG, srcm \, SG, tgtm \, SG, pe \, SG, ds \, SG, vcei \, SG)
\end{array}$$

function 10 **leftassoc** ($_ \rightarrow_{SG} _$)

$$\begin{array}{|l}
\hline
- \rightarrow_{SG} - : SGr \times SGr \rightarrow \mathbb{P} GrM \\
\hline
\forall SG_s, SG_t : SGr \bullet \\
SG_s \rightarrow_{SG} SG_t = \{fv : sg_Ns SG_s \rightarrow sg_Ns SG_t; fe : EsA SG_s \rightarrow EsA SG_t \mid \\
fv \circ (src_M^* SG_s) \subseteq src_M^* SG_t \circ fe \wedge fv \circ (tgt_M^* SG_s) \subseteq tgt_M^* SG_t \circ fe \\
\wedge fv \circ \preceq SG_s \subseteq \preceq SG_t \circ fv\}
\end{array}$$

function 10 **leftassoc** $(- \rightarrow_{SG} -)$

$$\begin{array}{|l}
\hline
- \rightarrow_{SG} - : SGr \times SGr \rightarrow \mathbb{P} GrM \\
\hline
\forall SG_s, SG_t : SGr \bullet \\
SG_s \rightarrow_{SG} SG_t = \{fv : sg_Ns SG_s \rightarrow sg_Ns SG_t; fe : EsA SG_s \rightarrow EsA SG_t \mid \\
fv \circ ((\text{dom } fe) \triangleleft (src_M^* SG_s)) \subseteq src_M^* SG_t \circ fe \wedge fv \circ ((\text{dom } fe) \triangleleft (tgt_M^* SG_s)) \subseteq tgt_M^* SG_t \circ fe \\
\wedge fv \circ \preceq SG_s \subseteq \preceq SG_t \circ fv\}
\end{array}$$

relation $(- \Rightarrow^{SG} -)$

$$\begin{array}{|l}
\hline
- \Rightarrow^{SG} - : \mathbb{P}((SGr \times GrM) \times SGr) \\
\hline
\forall SG_s, SG_t : SGr; m : GrM \bullet (SG_s, m) \Rightarrow^{SG} SG_t \Leftrightarrow m \in SG_s \rightarrow_{SG} SG_t
\end{array}$$

function 10 **leftassoc** $(- \rightarrow_{G2SG} -)$

$$\begin{array}{|l}
\hline
- \rightarrow_{G2SG} - : Gr \times SGr \rightarrow \mathbb{P} GrM \\
\hline
\forall G : Gr; SG : SGr \bullet G \rightarrow_{G2SG} SG = \{fv : Ns G \rightarrow sg_Ns SG; fe : Es G \rightarrow EsA SG \mid \\
fv \circ src G \subseteq src_M^* SG \circ fe \wedge fv \circ tgt G \subseteq tgt_M^* SG \circ fe\}
\end{array}$$

relation $(- \Rightarrow^{GwT} -)$

$$\begin{array}{|l}
\hline
- \Rightarrow^{GwT} - : (GrwT \Leftrightarrow SGr) \\
\hline
\forall GwT : GrwT; SG : SGr \bullet GwT \Rightarrow^{GwT} SG \Leftrightarrow GwT^T \in GwT^G \rightarrow_{G2SG} SG
\end{array}$$

relation $(- \sqsupseteq^{SG} -)$
relation $(- \sqsupseteq^{SG_0} -)$
relation $(- \sqsupseteq^{NT} -)$
relation $(- \sqsupseteq^{ET} -)$
relation $(- \sqsupseteq^{\mathcal{M}} -)$
relation $(- \sqsupseteq^{\mathcal{MCnts}} -)$

$$\begin{array}{|l}
- \sqsubseteq^{NT} -, - \sqsubseteq^{ET} - : \mathbb{P}((SGr \times GrM) \times SGr) \\
\hline
\forall SG_c, SG_a : SGr; m : GrM \bullet \\
(SG_c, m) \sqsubseteq^{NT} SG_a \Leftrightarrow \forall n : sg_Ns SG_c \bullet (nty SG_c) n \leq_{rNT} ((nty SG_a) \circ (fV m)) n \\
\forall SG_c, SG_a : SGr; m : GrM \bullet \\
(SG_c, m) \sqsubseteq^{ET} SG_a \Leftrightarrow \forall e : EsA SG_c \bullet (ety SG_c) e \leq_{ET} ((ety SG_a) \circ (fE m)) e
\end{array}$$

$$\begin{array}{|l}
sPEA, tPEA : SGr \rightarrow PEA \rightarrow (E \rightarrow V) \\
smfPEA, tmfPEA : SGr \rightarrow PEA \rightarrow (E \rightarrow Mult) \\
\hline
\forall SG : SGr; e : E \bullet sPEA SG (edg e) = sg_src SG \\
\forall SG : SGr; e : E \bullet sPEA SG (einv e) = sg_tgt SG \\
\forall SG : SGr; e : E \bullet tPEA SG (edg e) = sg_tgt SG \\
\forall SG : SGr; e : E \bullet tPEA SG (einv e) = sg_src SG \\
\forall SG : SGr; e : E \bullet smfPEA SG (edg e) = srcma SG \\
\forall SG : SGr; e : E \bullet smfPEA SG (einv e) = tgtm SG \\
\forall SG : SGr; e : E \bullet tmfPEA SG (edg e) = tgtm SG \\
\forall SG : SGr; e : E \bullet tmfPEA SG (einv e) = srcma SG
\end{array}$$

relation(*affectedPE* $_$)
relation(*affectedPECStart* $_$)
relation(*affectedPECEnd* $_$)
relation(*affectedPEAStart* $_$)
relation(*affectedPEAEnd* $_$)
relation(*affectedPEStart* $_$)
relation(*affectedPEEnd* $_$)

$$\begin{array}{l}
\text{affectedPECStart } _, \text{affectedPECEnd } _ : \mathbb{P}(SGr \times GrM \times SGr \times PEC \times E \times E) \\
\text{affectedPEAStart } _, \text{affectedPEAEnd } _ : \mathbb{P}(SGr \times GrM \times SGr \times PEA \times E \times E) \\
\hline
\forall SG_c, SG_a : SGr; m : GrM; pea : PEA; e, ie : E \bullet \text{affectedPEAStart}(SG_c, m, SG_a, pea, e, ie) \\
\quad \Leftrightarrow ie \in (fE \ m) \sim \llbracket \{ePEA \ pea\} \rrbracket \wedge (fV \ m)(sPEA \ SG_c \ pea \ ie) = sPEA \ SG_a \ pea \ e \\
\forall SG_c, SG_a : SGr; m : GrM; pea : PEA; e, ie : E \bullet \text{affectedPEAEnd}(SG_c, m, SG_a, pea, e, ie) \\
\quad \Leftrightarrow ie \in (fE \ m) \sim \llbracket \{ePEA \ pea\} \rrbracket \wedge (fV \ m)(tPEA \ SG_c \ pea \ ie) = tPEA \ SG_a \ pea \ e \\
\forall SG_a, SG_c : SGr; m : GrM; pea : PEA; e, ie : E \bullet \text{affectedPECStart}(SG_c, m, SG_a, eat \ pea, e, ie) \\
\quad \Leftrightarrow \text{affectedPEAStart}(SG_c, m, SG_a, pea, e, ie) \\
\forall SG_a, SG_c : SGr; m : GrM; pea : PEA; v : V; e, ie : E \bullet \\
\quad \text{affectedPECStart}(SG_c, m, SG_a, eresd \ (v, pea), e, ie) \\
\quad \Leftrightarrow \text{affectedPEAStart}(SG_c, m, SG_a, pea, e, ie) \\
\quad \wedge \exists v' : (fV \ m) \sim \llbracket \{v\} \rrbracket \bullet (sPEA \ SG_c \ pea \ ie, v') \in \preceq SG_c \\
\forall SG_a, SG_c : SGr; m : GrM; pea : PEA; e, ie : E \bullet \text{affectedPECEnd}(SG_c, m, SG_a, eat \ pea, e, ie) \\
\quad \Leftrightarrow \text{affectedPEAEnd}(SG_c, m, SG_a, pea, e, ie) \\
\forall SG_a, SG_c : SGr; m : GrM; pea : PEA; v : V; e, ie : E \bullet \\
\quad \text{affectedPECEnd}(SG_c, m, SG_a, eresd \ (v, pea), e, ie) \\
\quad \Leftrightarrow \text{affectedPEAEnd}(SG_c, m, SG_a, pea, e, ie) \\
\quad \wedge \exists v' : (fV \ m) \sim \llbracket \{v\} \rrbracket \bullet (sPEA \ SG_c \ pea \ ie, v') \in \preceq SG_c \\
\forall SG_a, SG_c : SGr; m : GrM; pea : PEA; v : V; e, ie : E \bullet \\
\quad \text{affectedPECEnd}(SG_c, m, SG_a, eresr \ (pea, v), e, ie) \\
\quad \Leftrightarrow \text{affectedPEAEnd}(SG_c, m, SG_a, pea, e, ie) \\
\quad \wedge \exists v' : (fV \ m) \sim \llbracket \{v\} \rrbracket \bullet (tPEA \ SG_c \ pea \ ie, v') \in \preceq SG_c
\end{array}$$

$affectedPEStart _, affectedPEEnd _ : \mathbb{P}(SGr \times GrM \times SGr \times PE \times E \times E)$
$affectedPE _ : \mathbb{P}(SGr \times GrM \times SGr \times PE \times E \times E \times E)$
$\forall SG_a, SG_c : SGr; m : GrM; pec : PEC; e, ie : E \bullet$ $affectedPEStart(SG_c, m, SG_a, ec\ pec, e, ie) \Leftrightarrow affectedPECStart(SG_c, m, SG_a, pec, e, ie)$
$\forall SG_a, SG_c : SGr; m : GrM; pec : PEC; pe : PE; e, ie : E \bullet$ $affectedPEStart(SG_c, m, SG_a, scmp(pec, pe), e, ie) \Leftrightarrow affectedPECStart(SG_c, m, SG_a, pec, e, ie)$
$\forall SG_a, SG_c : SGr; m : GrM; pec : PEC; e, ie : E \bullet$ $affectedPEEnd(SG_c, m, SG_a, ec\ pec, e, ie) \Leftrightarrow affectedPECEnd(SG_c, m, SG_a, pec, e, ie)$
$\forall SG_a, SG_c : SGr; m : GrM; pec : PEC; pe : PE; e, ie : E \bullet$ $affectedPEEnd(SG_c, m, SG_a, scmp(pec, pe), e, ie) \Leftrightarrow affectedPEEnd(SG_c, m, SG_a, pe, e, ie)$
$\forall SG_a, SG_c : SGr; m : GrM; pec : PEC; e, ie, ie' : E \bullet$ $affectedPE(SG_c, m, SG_a, ec\ pec, e, ie, ie')$ $\Leftrightarrow ie = ie' \wedge affectedPECStart(SG_c, m, SG_a, pec, e, ie)$ $\wedge affectedPECEnd(SG_c, m, SG_a, pec, e, ie)$
$\forall SG_a, SG_c : SGr; m : GrM; pec_1, pec_2 : PEC; e, ie, ie' : E \bullet$ $affectedPE(SG_c, m, SG_a, scmp(pec_1, ec\ pec_2), e, ie, ie')$ $\Leftrightarrow tPEA\ SG_c(endEA_C\ pec_1)\ ie = sPEA\ SG_c(startEA_C\ pec_2)\ ie'$ $\wedge affectedPECStart(SG_c, m, SG_a, pec_1, e, ie) \wedge affectedPECEnd(SG_c, m, SG_a, pec_2, e, ie')$
$\forall SG_a, SG_c : SGr; m : GrM; pec : PEC; pe : PE; e, ie, ie' : E \bullet$ $affectedPE(SG_c, m, SG_a, scmp(pec, pe), e, ie, ie')$ $\Leftrightarrow affectedPECStart(SG_c, m, SG_a, pec, e, ie)$ $\wedge \exists ie'' : E \mid tPEA\ SG_c(endEA_C\ pec)\ ie = sPEA\ SG_c(startEA\ pe)\ ie'' \bullet$ $affectedPE(SG_c, m, SG_a, pe, e, ie'', ie')$

relation(*caseMultsOk* $_$)

relation(*caseMandatoryT* $_$)

relation(*caseMandatoryS* $_$)

$caseMultsOk _ : \mathbb{P}(SGr \times GrM \times SGr \times E \times PEA \times PEA \times E \times E)$
$\forall SG_c, SG_a : SGr; m : GrM; peas : PEA; peae : PEA; e, ie, ie' : E \bullet$ $caseMultsOk(SG_c, m, SG_a, e, peas, peae, ie, ie')$ $\Leftrightarrow affectedPE(SG_c, m, SG_a, pe\ SG_a\ e, e, ie, ie')$ $\Rightarrow smfPEA\ SG_c\ peas\ ie \leq_{\mathcal{M}} srcma\ SG_a\ e \wedge tmfPEA\ SG_c\ peae\ ie' \leq_{\mathcal{M}} tgtm\ SG_a\ e$
$caseMandatoryT _ : \mathbb{P}(SGr \times GrM \times SGr \times E \times PEA \times E)$
$\forall SG_c, SG_a : SGr; m : GrM; peas : PEA; peae : PEA; e, ie : E \bullet$ $caseMandatoryT(SG_c, m, SG_a, e, peae, ie)$ $\Leftrightarrow (affectedPEStart(SG_c, m, SG_a, pe\ SG_a\ e, e, ie) \wedge mks\ 1 \leq_{\mathcal{M}} (tgtm\ SG_a)\ e)$ $\Rightarrow (fV\ m)(tPEA\ SG_c\ peae\ ie) = sg_tgt\ SG_a\ e$

$caseMandatoryS _ : \mathbb{P}(SGr \times GrM \times SGr \times E \times PEA \times E)$	$\begin{aligned} & \forall SG_c, SG_a : SGr; m : GrM; peas : PEA; e, ie : E \bullet \\ & \quad caseMandatoryS(SG_c, m, SG_a, e, peas, ie) \\ & \quad \Leftrightarrow (affectedPEEnd(SG_c, m, SG_a, pe\ SG_a\ e, e, ie) \wedge mks\ 1 \leq_{\mathcal{M}} (srcma\ SG_a)\ e) \\ & \quad \Rightarrow (fV\ m)(sPEA\ SG_c\ peas\ ie) = sg_src\ SG_a\ e \end{aligned}$
$_ \sqsubseteq_{\mathcal{MCnts}} _ : \mathbb{P}((SGr \times GrM) \times SGr)$ $_ \sqsubseteq^{\mathcal{M}} _ : \mathbb{P}((SGr \times GrM) \times SGr)$	$\begin{aligned} & \forall SG_c, SG_a : SGr; m : GrM \bullet \\ & \quad (SG_c, m) \sqsubseteq^{\mathcal{M}} SG_a \Leftrightarrow \forall e : EsA\ SG_c \bullet (srcma\ SG_c)\ e \leq_{\mathcal{M}} ((srcma\ SG_a) \circ (fE\ m))\ e \\ & \quad \quad \wedge (tgtm\ SG_c)\ e \leq_{\mathcal{M}} ((tgtm\ SG_a) \circ (fE\ m))\ e \\ & \forall SG_c, SG_a : SGr; m : GrM \bullet (SG_c, m) \sqsubseteq_{\mathcal{MCnts}} SG_a \Leftrightarrow \forall e : EsD\ SG_a \bullet \\ & \quad \exists peas : PEA; peae : PEA \mid peas = startEA(pe\ SG_a\ e) \wedge peae = endEA(pe\ SG_a\ e) \bullet \\ & \quad \quad \forall e' : (fE\ m) \sim \langle \{ePEA\ peas\} \rangle; e'' : (fE\ m) \sim \langle \{ePEA\ peae\} \rangle \bullet \\ & \quad \quad caseMultsOk(SG_c, m, SG_a, e, peas, peae, e', e'') \\ & \quad \quad \wedge (caseMandatoryT(SG_c, m, SG_a, e, peae, e') \vee caseMandatoryS(SG_c, m, SG_a, e, peas, e'')) \end{aligned}$
$_ \sqsubseteq^{SG} _, _ \sqsubseteq^{SG_0} _ : \mathbb{P}((SGr \times GrM) \times SGr)$	$\begin{aligned} & \forall SG_c, SG_a : SGr; m : GrM \bullet \\ & \quad (SG_c, m) \sqsubseteq^{SG_0} SG_a \Leftrightarrow (SG_c, m) \sqsubseteq^{NT} SG_a \wedge (SG_c, m) \sqsubseteq^{ET} SG_a \wedge (SG_c, m) \sqsubseteq^{\mathcal{M}} SG_a \\ & \quad \quad \wedge (SG_c, m) \sqsubseteq_{\mathcal{MCnts}} SG_a \\ & \forall SG_c, SG_a : SGr; m : GrM \bullet \\ & \quad (SG_c, m) \sqsubseteq^{SG} SG_a \Leftrightarrow m \in SG_c \rightarrow_{SG} SG_a \wedge (SG_c, m) \sqsubseteq^{SG_0} SG_a \end{aligned}$
$ins : GrM \times SGr \times \mathbb{P}\ V \rightarrow \mathbb{P}\ V$ $ies : GrM \times \mathbb{P}\ E \rightarrow \mathbb{P}\ E$	$\begin{aligned} & \forall m : GrM; SG : SGr; mns : \mathbb{P}\ V \bullet ins\ (m, SG, mns) = (fV\ m) \sim \langle (\prec\ SG) \sim \langle mns \rangle \rangle \\ & \forall m : GrM; mes : \mathbb{P}\ E \bullet ies\ (m, mes) = (fE\ m) \sim \langle mes \rangle \end{aligned}$
$igRMEs : GrwT \times \mathbb{P}\ E \rightarrow Gr$	$\forall GwT : GrwT; mes : \mathbb{P}\ E \bullet igRMEs(GwT, mes) = GwT^G \bowtie_{Es} ies\ (GwT^T, mes)$

relation($_ \sqsubseteq^{SG} _$)
relation($_ \sqsubseteq^{SG_0} _$)
relation($_ \sqsubseteq_{AEs} _$)
relation($_ OkRefinedIn _$)
relation($_ \sqsubseteq_{ANNs} _$)

$$\begin{array}{|l}
\hline
- \sqsupset_{ANNs} - : \mathbb{P}(GrM \times SGr) \\
\hline
\forall SG_a : SGr; m : GrM \bullet \\
m \sqsupset_{ANNs} SG_a \Leftrightarrow \forall nn : NsTy SG_a \{nnrml\} \bullet (\preceq SG_a) \parallel \{nn\} \parallel \cap \text{ran}(fV m) = \emptyset
\end{array}$$

$$\begin{array}{|l}
\hline
OkRefinedIn : \mathbb{P}((SGr \times E) \times (SGr \times GrM)) \\
- \sqsupset_{AEs} - : \mathbb{P}((SGr \times GrM) \times SGr) \\
\hline
\forall SG_c, SG_a : SGr; m : GrM; ae : E \bullet \\
(SG_a, ae)OkRefinedIn(SG_c, m) \Leftrightarrow \\
\exists r : V \leftrightarrow V; s, t : \mathbb{P} V \mid r = (\preceq SG_c) \circ igRMEs((gr SG_c, m), \{ae\})^{\leftrightarrow} \circ (\preceq SG_c) \sim \\
\wedge s = ins(m, SG_a, sg_src SG_a \parallel \{ae\} \parallel) \setminus ((NsEther SG_c) \setminus \text{dom } r) \\
\wedge t = ins(m, SG_a, sg_tgt SG_a \parallel \{ae\} \parallel) \setminus ((NsEther SG_c) \setminus \text{ran } r) \\
\bullet r \in s \leftrightarrow t \wedge nty SG_a \parallel (gr SG_a) \circ \rightarrow_{Ns} \{ae\} \parallel \neq \emptyset \Rightarrow r \neq \emptyset \\
\forall SG_c, SG_a : SGr; m : GrM \bullet \\
(SG_c, m) \sqsupset_{AEs} SG_a \Leftrightarrow \forall e : (EsA SG_a) \bullet (SG_a, e)OkRefinedIn(SG_c, m)
\end{array}$$

$$\begin{array}{|l}
\hline
- \sqsupset^{SG} -, - \sqsupset^{SG_0} - : \mathbb{P}((SGr \times GrM) \times SGr) \\
\hline
\forall SG_c, SG_a : SGr; m : GrM \bullet \\
(SG_c, m) \sqsupset^{SG_0} SG_a \Leftrightarrow (SG_c, m) \sqsupseteq^{SG_0} SG_a \wedge m \sqsupset_{ANNs} SG_a \wedge (SG_c, m) \sqsupset_{AEs} SG_a \\
\forall SG_c, SG_a : SGr; m : GrM \bullet \\
(SG_c, m) \sqsupset^{SG} SG_a \Leftrightarrow m \in SG_c \rightarrow_{SG} SG_a \wedge (SG_c, m) \sqsupset^{SG_0} SG_a
\end{array}$$

relation($- \Vdash_{\mathcal{M}} -$)
relation($- \Vdash_{NT} -$)
relation($- \Vdash_{ET} -$)
relation($- \Vdash_{SG} -$)

$$\begin{array}{|l}
\hline
- \Vdash_{\mathcal{M}} - : \mathbb{P}((SGr \times GrM) \times SGr) \\
\hline
\forall SG_s, SG_t : SGr; m : GrM \bullet \\
(SG_s, m) \Vdash_{\mathcal{M}} SG_t \Leftrightarrow \forall ep : fE m \bullet (srcma SG_s) (first ep) \leq_{\mathcal{M}} (srcma SG_t) (second ep) \\
\wedge (tgtm SG_s) (first ep) \leq_{\mathcal{M}} (tgtm SG_t) (second ep)
\end{array}$$

$$\begin{array}{|l}
\hline
- \Vdash_{NT} - : \mathbb{P}(GrM \times SGr) \\
- \Vdash_{ET} - : \mathbb{P}((SGr \times GrM) \times SGr) \\
\hline
\forall SG_t : SGr; m : GrM \bullet \\
m \Vdash_{NT} SG_t \Leftrightarrow (nty SG_t) \parallel \text{ran}(fV m) \parallel \subseteq \{nnrml, nabst, nvirt\} \\
\forall SG_s, SG_t : SGr; m : GrM \bullet \\
(SG_s, m) \Vdash_{ET} SG_t \Leftrightarrow \text{dom}(fE m) \triangleleft (ety SG_s) = (ety SG_t) \circ (fE m)
\end{array}$$

$\frac{}{_ \Vdash^{SG} _ : \mathbb{P}((SGr \times GrM) \times SGr)}$	$\frac{}{\forall SG_s, SG_t : SGr; m : GrM \bullet (SG_s, m) \Vdash^{SG} SG_t \Leftrightarrow m \in SG_s \rightarrow_{SG} SG_t \wedge (SG_s, m) \Vdash_{\mathcal{M}} SG_t \wedge m \Vdash_{NT} SG_t \wedge (SG_s, m) \Vdash_{ET} SG_t}$
$\frac{}{rPEA : GrwT \times SGr \times PEA \rightarrow (V \leftrightarrow V)}$	$\frac{}{\forall GwT : GrwT; SG : SGr; e : E \bullet rPEA(GwT, SG, edg\ e) = (GwT^G \bowtie_{Es} ies(GwT^T, \{e\})) \leftrightarrow \sim}$ $\forall GwT : GrwT; SG : SGr; e : E \bullet rPEA(GwT, SG, einv\ e) = rPEA(GwT, SG, edg\ e) \sim$
$\frac{}{rPEC : GrwT \times SGr \times PEC \rightarrow (V \leftrightarrow V)}$ $\frac{}{rPE : GrwT \times SGr \times PE \rightarrow (V \leftrightarrow V)}$	$\forall GwT : GrwT; SG : SGr; pea : PEA \bullet rPEC(GwT, SG, eat\ pea) = rPEA(GwT, SG, pea)$ $\forall GwT : GrwT; SG : SGr; v : V; pea : PEA \bullet$ $rPEC(GwT, SG, eresd(v, pea)) = ins(GwT^T, SG, \{v\}) \triangleleft rPEA(GwT, SG, pea)$ $\forall GwT : GrwT; SG : SGr; v : V; pea : PEA \bullet$ $rPEC(GwT, SG, eresr(pea, v)) = rPEA(GwT, SG, pea) \triangleright ins(GwT^T, SG, \{v\})$ $\forall GwT : GrwT; SG : SGr; pec : PEC \bullet$ $rPE(GwT, SG, ec\ pec) = rPEC(GwT, SG, pec)$ $\forall GwT : GrwT; SG : SGr; pec : PEC; pe : PE \bullet$ $rPE(GwT, SG, scmp(pec, pe)) = rPEC(GwT, SG, pec) \mathbin{\circ} rPE(GwT, SG, pe)$
$\frac{}{ape : SGr \rightarrow E \leftrightarrow PE}$	$\forall SG : SGr \bullet ape\ SG = (\lambda e : E \mid e \in EsM\ SG \bullet (ec \circ eat \circ edg)\ e) \oplus pe\ SG$
$\frac{}{src_{PEA}^* : SGr \times PEA \rightarrow E \leftrightarrow V}$ $\frac{}{src_{PEC}^* : SGr \times PEC \rightarrow E \leftrightarrow V}$ $\frac{}{src_{PE}^* : SGr \times PE \rightarrow E \leftrightarrow V}$	$\forall SG : SGr; e : E \bullet src_{PEA}^*(SG, edg\ e) = src^*\ SG$ $\forall SG : SGr; e : E \bullet src_{PEA}^*(SG, einv\ e) = tgt^*\ SG$ $\forall SG : SGr; pea : PEA \bullet src_{PEC}^*(SG, eat\ pea) = src_{PEA}^*(SG, pea)$ $\forall SG : SGr; v : V; pea : PEA \bullet src_{PEC}^*(SG, eresd(v, pea)) = src_{PEA}^*(SG, pea) \triangleright \{v\}$ $\forall SG : SGr; v : V; pea : PEA \bullet src_{PEC}^*(SG, eresr(pea, v)) = src_{PEA}^*(SG, pea)$ $\forall SG : SGr; v : V; pec : PEC; pe : PE \bullet src_{PE}^*(SG, (scmp(pec, pe))) = src_{PEC}^*(SG, pec)$

$tgt_{PEA}^* : SGr \times PEA \rightarrow E \leftrightarrow V$ $tgt_{PEC}^* : SGr \times PEC \rightarrow E \leftrightarrow V$ $tgt_{PE}^* : SGr \times PE \rightarrow E \leftrightarrow V$
$\forall SG : SGr; e : E \bullet tgt_{PEA}^*(SG, edg\ e) = tgt^*\ SG$ $\forall SG : SGr; e : E \bullet tgt_{PEA}^*(SG, einv\ e) = src^*\ SG$ $\forall SG : SGr; pea : PEA \bullet tgt_{PEC}^*(SG, eat\ pea) = tgt_{PEA}^*(SG, pea)$ $\forall SG : SGr; v : V; pea : PEA \bullet tgt_{PEC}^*(SG, eresd\ (v, pea)) = tgt_{PEA}^*(SG, pea)$ $\forall SG : SGr; v : V; pea : PEA \bullet tgt_{PEC}^*(SG, eresr\ (pea, v)) = tgt_{PEA}^*(SG, pea) \triangleright \{v\}$ $\forall SG : SGr; v : V; pec : PEC \bullet tgt_{PE}^*(SG, ec\ pec) = tgt_{PEC}^*(SG, pec)$ $\forall SG : SGr; v : V; pec : PEC; pe : PE \bullet tgt_{PE}^*(SG, (scmp\ (pec, pe))) = tgt_{PE}^*(SG, pe)$

relation($_ \ni^{SG} _$)
relation($_ \ni_{\mathcal{M}} _$)
relation($_ \ni_{FI} _$)
relation($_ \ni_{PNS} _$)
relation($_ \ni_{Cnts} _$)
relation($_ MEMOk _$)

$rMEMOk : SGr \times E \times GrwT \rightarrow V \leftrightarrow V$
$\forall SG : SGr; me : E; GwT : GrwT; s, t : \mathbb{P}\ V \mid s = ins(GwT^T, SG, src^*\ SG \parallel \{me\}) \parallel$ $\wedge t = ins(GwT^T, SG, tgt^*\ SG \parallel \{me\}) \bullet$ $rMEMOk(SG, me, GwT) = s \triangleleft rPE\ (GwT, SG, ape\ SG\ me) \triangleright t$

$src^*MEMOk : SGr \times E \rightarrow E \leftrightarrow V$
$\forall SG : SGr; me : E \bullet src^*MEMOk(SG, me) = src_{PE}^*(SG, ape\ SG\ me) \triangleright (src^*\ SG) \parallel \{me\} \parallel$

$tgt^*MEMOk : SGr \times E \rightarrow E \leftrightarrow V$
$\forall SG : SGr; me : E \bullet tgt^*MEMOk(SG, me) = tgt_{PE}^*(SG, ape\ SG\ me) \triangleright (tgt^*\ SG) \parallel \{me\} \parallel$

$multComp : Mult \times Mult \rightarrow Mult$
$\forall m_1, m_2 : Mult \mid m_1 \in MultMany \vee m_2 \in MultMany \bullet multComp(m_1, m_2) = mmanys$
$\forall m_1, m_2 : Mult \mid m_1 = mks\ 0 \vee m_2 = mks\ 0 \bullet multComp(m_1, m_2) = mks\ 0$
$\forall m_1, m_2 : Mult \mid m_2 = mks\ 1 \bullet multComp(m_1, m_2) = m_1$
$\forall m_1, m_2 : Mult \mid m_1 = mks\ 1 \bullet multComp(m_1, m_2) = m_2$
$\forall m_1, m_2 : Mult \mid m_1 = mopts \bullet multComp(m_1, m_2) = mks\ 0 \cup m_2$
$\forall m_1, m_2 : Mult \mid m_2 = mopts \bullet multComp(m_1, m_2) = mks\ 0 \cup m_1$
$\forall m_1, m_2 : MultRange \bullet$ $multComp(m_1, m_2) = \{(\text{the } m_1) *_{mr} (\text{the } m_2)\}$
$\forall m_1 : Mult; mc : MultC; m_2 : MultEither \bullet$ $multComp(m_1, \{mc\} \cup m_2) = multComp(m_1, \{mc\}) \cup multComp(m_1, m_2)$
$\forall m_1 : MultEither; mc : MultC; m_2 : Mult \bullet$ $multComp(\{mc\} \cup m_1, m_2) = multComp(\{mc\}, m_2) \cup multComp(m_1, m_2)$
$smPEA : SGr \rightarrow PEA \rightarrow Mult$ $smPEC : SGr \rightarrow PEC \rightarrow Mult$ $smPE : SGr \rightarrow PE \rightarrow Mult$
$\forall SG : SGr; e : E \bullet smPEA\ SG\ (edg\ e) = srcma\ SG\ e$
$\forall SG : SGr; e : E \bullet smPEA\ SG\ (einv\ e) = tgtm\ SG\ e$
$\forall SG : SGr; pea : PEA \bullet smPEC\ SG\ (eat\ pea) = smPEA\ SG\ pea$
$\forall SG : SGr; v : V; pea : PEA \bullet smPEC\ SG\ (eresd\ (v, pea)) = smPEA\ SG\ pea$
$\forall SG : SGr; v : V; pea : PEA \bullet smPEC\ SG\ (eresr\ (pea, v)) = smPEA\ SG\ pea$
$\forall SG : SGr; pec : PEC \bullet smPE\ SG\ (ec\ pec) = smPEC\ SG\ pec$
$\forall SG : SGr; pec : PEC; pe : PE \bullet smPE\ SG\ (scmp\ (pec, pe)) = multComp(smPEC\ SG\ pec, smPE\ SG\ pe)$
$tmPEA : SGr \rightarrow PEA \rightarrow Mult$ $tmPEC : SGr \rightarrow PEC \rightarrow Mult$ $tmPE : SGr \rightarrow PE \rightarrow Mult$
$\forall SG : SGr; e : E \bullet tmPEA\ SG\ (edg\ e) = tgtm\ SG\ e$
$\forall SG : SGr; e : E \bullet tmPEA\ SG\ (einv\ e) = srcma\ SG\ e$
$\forall SG : SGr; pea : PEA \bullet tmPEC\ SG\ (eat\ pea) = tmPEA\ SG\ pea$
$\forall SG : SGr; v : V; pea : PEA \bullet tmPEC\ SG\ (eresd\ (v, pea)) = tmPEA\ SG\ pea$
$\forall SG : SGr; v : V; pea : PEA \bullet tmPEC\ SG\ (eresr\ (pea, v)) = tmPEA\ SG\ pea$
$\forall SG : SGr; pec : PEC \bullet tmPE\ SG\ (ec\ pec) = tmPEC\ SG\ pec$
$\forall SG : SGr; pec : PEC; pe : PE \bullet tmPE\ SG\ (scmp\ (pec, pe)) = multComp(tmPEC\ SG\ pec, tmPE\ SG\ pe)$

$_MEMOk_ : \mathbb{P}((SGr \times E) \times GrwT)$
$\begin{aligned} \forall GrwT : GrwT; SG : SGr; me : E \bullet (SG, me) MEMOk GrwT \Leftrightarrow \\ \exists r : V \leftrightarrow V; s, t : \mathbb{P} V \mid r = rMEMOk(SG, me, GrwT) \\ \wedge s = ins(GrwT^T, SG, src^* MEMOk(SG, me) \parallel \{rsrcPE(ape SG me)\}) \\ \wedge t = ins(GrwT^T, SG, tgt^* MEMOk(SG, me) \parallel \{rtgtPE(ape SG me)\}) \\ \bullet rMOk(r, s, t, smPE SG(ape SG me), tmPE SG(ape SG me)) \end{aligned}$

relation(INumbersOk₋)

relation(INsOk₋)

relation(IZsOk₋)

relation(IRsOk₋)

$INsOk_ , IZsOk_ , IRsOk_ , INumbersOk_ : \mathbb{P} GrwT$
$\begin{aligned} \forall GrwT : GrwT \bullet INsOk GrwT \Leftrightarrow \\ \forall nnat : (fV(GrwT^T)) \sim (\{nNatS\}) \bullet \exists n : \mathbb{N} \bullet n \in to\mathbb{Z} nnat \\ \forall GrwT : GrwT \bullet IZsOk GrwT \Leftrightarrow \\ \forall nint : (fV(GrwT^T)) \sim (\{nIntS\}) \bullet \exists n : \mathbb{Z} \bullet n \in to\mathbb{Z} nint \\ \forall GrwT : GrwT \bullet IRsOk GrwT \Leftrightarrow \\ \forall nreal : (fV(GrwT^T)) \sim (\{nRealS\}) \bullet \exists x : \mathbb{R} \bullet x \in to\mathbb{R} nreal \\ \forall SG : SGr; GrwT : GrwT \bullet INumbersOk GrwT \Leftrightarrow \\ INsOk GrwT \wedge IZsOk GrwT \wedge IRsOk GrwT \end{aligned}$

relation(satisfiesCnt₋)

relation(satisfiesVCEECnt₋)

relation(satisfiesVCENCnt₋)

relation(IVCEsOk₋)

$toNum : SGr \times V \times V \rightarrow \text{opt}[\mathbb{A}]$
$\begin{aligned} \forall SG : SGr; ns, nt : V \bullet toNum(SG, ns, nt) = \\ \text{if}(nt, nNatS) \in (\preceq SG) \vee (nt, nIntS) \in (\preceq SG) \text{ then } to\mathbb{Z} ns \\ \text{ else if}(nt, nRealS) \in (\preceq SG) \text{ then } to\mathbb{R} ns \text{ else } \emptyset \end{aligned}$

$rOp : SGVCEOP \rightarrow \mathbb{A} \leftrightarrow \mathbb{A}$
$\begin{aligned} rOp eq &= \{n_1, n_2 : \mathbb{A} \mid n_1 = n_2\} \\ rOp neq &= (- \neq -) \\ rOp leq &= (- \leq -) \\ rOp geq &= (- \geq -) \\ rOp lt &= (- < -) \\ rOp gt &= (- > -) \end{aligned}$

$\text{satisfiesCnt}_- : \mathbb{P}(SGr \times GrM \times V \times SGVCEOP \times V)$ $\text{satisfiesVCEECnt}_- : \mathbb{P}(SGr \times GrwT \times E)$ $\text{satisfiesVCENCnt}_- : \mathbb{P}(SGr \times GrwT \times E)$ $\text{IVCEsOk}_- : \mathbb{P}(SGr \times GrwT)$	
$\forall SG : SGr; t : GrM; op : SGVCEOP; ns, nt : V \bullet \text{satisfiesCnt}(SG, t, ns, op, nt) \Leftrightarrow$ $ns \in \text{dom}(fV\ t)$ $\wedge (\exists n_1, n_2 : \mathbb{A} \mid n_1 \in \text{toNum}(SG, ns, (fV\ t)\ ns) \wedge n_2 \in \text{toNum}(SG, nt, nt) \bullet (n_1, n_2) \in rOp\ op$ $\vee (op = eq \wedge (fV\ t)\ ns = nt \wedge nt \in NsVa\ SG))$	
$\forall SG : SGr; GwT : GrwT; vce : E \bullet \text{satisfiesVCEECnt}(SG, GwT, vce) \Leftrightarrow$ $\forall ie : fE\ (GwT^T) \sim \llbracket (second \circ (vcei\ SG))\ vce \rrbracket \mid fV\ (GwT^T)(src\ (GwT^G)\ ie) = (sg_src\ SG\ vce) \bullet$ $\text{satisfiesCnt}(SG, GwT^T, tgt\ (GwT^G)\ ie, (first \circ (vcei\ SG))\ vce, sg_tgt\ SG\ vce)$	
$\forall SG : SGr; GwT : GrwT; vce : E \bullet \text{satisfiesVCENCnt}(SG, GwT, vce) \Leftrightarrow$ $\forall in : fV\ (GwT^T) \sim \llbracket \{sg_src\ SG\ vce\} \rrbracket \bullet$ $\text{satisfiesCnt}(SG, GwT^T, in, (first \circ (vcei\ SG))\ vce, sg_tgt\ SG\ vce)$	
$\forall SG : SGr; GwT : GrwT \bullet \text{IVCEsOk}(SG, GwT) \Leftrightarrow \forall vce : EsVCnt\ SG \bullet$ $\text{isVCEECnt}(SG, vce) \Rightarrow \text{satisfiesVCEECnt}(SG, GwT, vce)$ $\wedge \text{isVCENCnt}(SG, vce) \Rightarrow \text{satisfiesVCENCnt}(SG, GwT, vce)$	
$_ \ni_{\mathcal{M}} _ : GrwT \leftrightarrow SGr$ $_ \ni_{FI} _ : GrwT \leftrightarrow SGr$ $_ \ni_{PNS} _ : GrwT \leftrightarrow SGr$ $_ \ni_{Cnts} _ : GrwT \leftrightarrow SGr$	
$\forall GwT : GrwT; SG : SGr \bullet$ $GwT \ni_{\mathcal{M}} SG \Leftrightarrow \forall me : EsM\ SG \bullet (SG, me) MEMOk\ GwT$	
$\forall GwT : GrwT; SG : SGr \bullet$ $GwT \ni_{FI} SG \Leftrightarrow (fV\ (GwT^T)) \sim \llbracket NsEther\ SG \rrbracket = \emptyset$	
$\forall GwT : GrwT; SG : SGr \bullet$ $GwT \ni_{PNS} SG \Leftrightarrow \forall ec : EsTy\ SG \{ecomp\ dbi, ecomp\ duni\} \bullet igRMEs(GwT, \{ec\})^{**} \in \text{injrel}$	
$\forall GwT : GrwT; SG : SGr \bullet$ $GwT \ni_{Cnts} SG \Leftrightarrow \text{INumbersOk}\ GwT \wedge \text{IVCEsOk}(SG, GwT)$	
$_ \ni^{SG} _ : GrwT \leftrightarrow SGr$	
$\forall GwT : GrwT; SG : SGr \bullet GwT \ni^{SG} SG \Leftrightarrow GwT \ni^{GwT} SG \wedge GwT \ni_{\mathcal{M}} SG$ $\wedge GwT \ni_{FI} SG \wedge GwT \ni_{PNS} SG \wedge GwT \ni_{Cnts} SG$	

8 Fragments

section *Fragmenta_Frs* **parents** *standard_toolkit, Fragmenta_Generics, Fragmenta_SGs, Fragmenta_GrswT, Fragmenta_GrswET*

$$Fr_0 == \{SG : SGr_0; esr : \mathbb{P} E; sr, tr : E \leftrightarrow V; et : GrM \mid esr \cap (sg_Es SG) = \emptyset \\ \wedge sr \in esr \mapsto (NsP SG) \wedge tr \in esr \rightarrow V \wedge domg et \subseteq_p (NsN SG, EsA SG)\}$$

$\begin{array}{l} fSG : Fr_0 \rightarrow SGr \\ EsR : Fr_0 \rightarrow \mathbb{P} E \\ srcR, tgtR : Fr_0 \rightarrow E \leftrightarrow V \\ fet : Fr_0 \rightarrow GrM \end{array}$	$\begin{array}{l} \forall SG : SGr; esr : \mathbb{P} E; sr, tr : E \leftrightarrow V; et : GrM \bullet fSG(SG, esr, sr, tr, et) = SG \\ \forall SG : SGr; esr : \mathbb{P} E; sr, tr : E \leftrightarrow V; et : GrM \bullet EsR(SG, esr, sr, tr, et) = esr \\ \forall SG : SGr; esr : \mathbb{P} E; sr, tr : E \leftrightarrow V; et : GrM \bullet srcR(SG, esr, sr, tr, et) = sr \\ \forall SG : SGr; esr : \mathbb{P} E; sr, tr : E \leftrightarrow V; et : GrM \bullet tgtR(SG, esr, sr, tr, et) = tr \\ \forall SG : SGr; esr : \mathbb{P} E; sr, tr : E \leftrightarrow V; et : GrM \bullet fet(SG, esr, sr, tr, et) = et \end{array}$
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$\begin{array}{l} fLEs, fEs, fEsA : Fr_0 \rightarrow \mathbb{P} E \\ fLNs, fRNs, fNs : Fr_0 \rightarrow \mathbb{P} V \\ srcF, tgtF : Fr_0 \rightarrow E \leftrightarrow V \end{array}$	$\begin{array}{l} fLEs = (sg_Es \circ fSG) \\ \forall F : Fr_0 \bullet fEs F = fLEs F \cup EsR F \\ fEsA = EsA \circ fSG \\ fLNs = sg_Ns \circ fSG \\ fRNs = ran \circ tgtR \\ \forall F : Fr_0 \bullet fNs F = fLNs F \cup fRNs F \\ \forall F : Fr_0 \bullet srcF F = (sg_src \circ fSG) F \cup srcR F \\ \forall F : Fr_0 \bullet tgtF F = (sg_tgt \circ fSG) F \cup tgtR F \end{array}$
--	---

$\begin{array}{l} \overset{G}{\rightsquigarrow} : Fr_0 \rightarrow Gr \\ \rightsquigarrow : Fr_0 \rightarrow V \leftrightarrow V \end{array}$	$\begin{array}{l} \forall F : Fr_0 \bullet \overset{G}{\rightsquigarrow} F = ((NsP \circ fSG) F \cup fRNs F, EsR F, srcR F, tgtR F) \\ \forall F : Fr_0 \bullet \rightsquigarrow F = (\overset{G}{\rightsquigarrow} F)^{\leftrightarrow} \end{array}$
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function 10 leftassoc $(_ \cup_F _)$

$$\begin{array}{|l}
\varnothing_F : Fr_0 \\
- \cup_F - : Fr_0 \times Fr_0 \rightarrow Fr_0 \\
\bigcup_F : \mathbb{P} Fr_0 \rightarrow Fr_0 \\
\hline
\varnothing_F = (\varnothing_{SG}, \varnothing, \varnothing, \varnothing, \varnothing_{GM}) \\
\forall F_1, F_2 : Fr_0 \bullet F_1 \cup_F F_2 = \\
(fSG F_1 \cup_{SG} fSG F_2, EsR F_1 \cup EsR F_2, srcR F_1 \cup srcR F_2, tgtR F_1 \cup tgtR F_2, fet F_1 \cup_{GM} fet F_2) \\
\bigcup_F \{ \} = \varnothing_F \\
\forall F : Fr_0; Fs : \mathbb{P} Fr_0 \bullet \bigcup_F (\{F\} \cup Fs) = F \cup_F (\bigcup_F Fs)
\end{array}$$

$$\begin{array}{|l}
\rightsquigarrow : Fr_0 \rightarrowtail V \rightarrowtail V \\
\bigodot^{SG} : Fr_0 \rightarrowtail SGr \\
rEsR : Fr_0 \rightarrowtail \mathbb{P} E \\
\bigodot : Fr_0 \rightarrowtail Fr_0 \\
\hline
\forall F : Fr_0 \bullet \rightsquigarrow F = (\rightsquigarrow F) \triangleright (fLNs F) \\
\forall F : Fr_0 \bullet \bigodot^{SG} F = (fSG F) \odot^{SG} (\rightsquigarrow F) \\
\forall F : Fr_0 \bullet rEsR F = \text{dom}((srcR F) \triangleright \text{dom}(\rightsquigarrow F)) \\
\forall F : Fr_0 \bullet \bigodot F = ((\bigodot^{SG} F, rEsR F, (rEsR F) \triangleleft (srcR F), (rEsR F) \triangleleft (tgtR F), fet F)
\end{array}$$

$$\begin{aligned}
Fr_a &== \{F : Fr_0 \mid \bigodot(\rightsquigarrow^G F)\} \\
Fr &== \{F : Fr_a \mid \bigodot^{SG} F \in SGr\}
\end{aligned}$$

relation(refsLocal₋)

$$\begin{array}{|l}
\text{refsLocal}_- : \mathbb{P} Fr_0 \\
\hline
\forall F : Fr_0 \bullet \text{refsLocal } F \Leftrightarrow fRNs F \subseteq fLNs F
\end{array}$$

$$TFr == \{F : Fr_a \mid \text{refsLocal } F \wedge \bigodot^{SG} F \in TSGr\}$$

relation($\boxminus$ ₋)
relation($\boxplus$ ₋)

$$\begin{array}{|l}
\boxminus_- : Fr \leftrightarrow Fr \\
\hline
\forall F_1, F_2 : Fr \bullet \boxminus(F_1, F_2) \Leftrightarrow fLNs F_1 \cap fLNs F_2 = \varnothing \wedge fEs F_1 \cap fEs F_2 = \varnothing
\end{array}$$

$[I]$
$\boxplus - : \mathbb{P}(I \rightarrow Fr)$
$\forall Fs : I \rightarrow Fr \bullet \boxplus Fs \Leftrightarrow \forall i, j : \text{dom } Fs \mid i \neq j \bullet \boxplus (Fs\ i, Fs\ j)$

relation $(- \subseteq^{rs} -)$
relation $(- \Rightarrow -)$

$- \subseteq^{rs} - : Fr \leftrightarrow Fr$
$- \Rightarrow - : Fr \leftrightarrow Fr$
$\forall F_1, F_2 : Fr \bullet F_1 \subseteq^{rs} F_2 \Leftrightarrow \text{ran}(tgtR\ F_1) \cap fLNs\ F_2 \neq \emptyset$
$\forall F_1, F_2 : Fr \bullet F_1 \Rightarrow F_2 \Leftrightarrow F_1 \subseteq^{rs} F_2 \wedge \neg (F_2 \subseteq^{rs} F_1)$

function 10 leftassoc $(- \Rightarrow_{\bullet} -)$

$- \Rightarrow_{\bullet} - : GrM \times (Fr \times Fr) \rightarrow GrM$
$\forall m : GrM; F_s, F_t : Fr_0 \bullet$ $m \Rightarrow_{\bullet} (F_s, F_t) = (((\leadsto F_s)^{\oplus} \boxplus (fLNs\ F_s)) \sim_{\circ} (fV\ m) \circ ((\leadsto F_t)^{\oplus} \boxplus (fLNs\ F_t)), fE\ m)$

function 1 leftassoc $(- \rightarrow_F -)$

$- \rightarrow_F - : Fr \times Fr \rightarrow \mathbb{P}\ GrM$
$\forall F_s, F_t : Fr \bullet F_s \rightarrow_F F_t = \{fv : fLNs\ F_s \rightarrow fLNs\ F_t; fe : fEsA\ F_s \rightarrow fEsA\ F_t \mid$ $(\bullet^{SG} F_s, (fv, fe) \Rightarrow_{\bullet} (F_s, F_t)) \Rightarrow^{SG} \bullet^{SG} F_t\}$

relation $(- \Rightarrow^F -)$

$- \Rightarrow^F - : (Fr \times GrM) \leftrightarrow Fr$
$\forall m : GrM; F_s, F_t : Fr_0 \bullet (F_s, m) \Rightarrow^F F_t \Leftrightarrow m \in F_s \rightarrow_F F_t$

relation($_{-} \sqsupseteq^F _{-}$)

$$\frac{_{-} \sqsupseteq^F _{-} : (Fr \times GrM) \leftrightarrow Fr}{\forall F_c, F_a : Fr_0; m : GrM \bullet (F_c, m) \sqsupseteq^F F_a \Leftrightarrow (F_c, m) \Rightarrow^F F_a \\ \wedge ((\bigodot^{SG} F_c, m \Rightarrow_{\bullet} (F_c, F_a)) \sqsupseteq^{SG_0} (\bigodot^{SG} F_a))}$$

relation($_{-} \sqsubset^F _{-}$)

$$\frac{_{-} \sqsubset^F _{-} : (Fr \times GrM) \leftrightarrow Fr}{\forall F_c, F_a : Fr_0; m : GrM \bullet (F_c, m) \sqsubset^F F_a \Leftrightarrow (F_c, m) \Rightarrow^F F_a \wedge \{F_a, F_c\} \subseteq Fr \\ \wedge ((\bigodot^{SG} F_c, m \Rightarrow_{\bullet} (F_c, F_a)) \sqsubset^{SG_0} (\bigodot^{SG} F_a))}$$

relation($_{-} \ni^F _{-}$)

$$\frac{_{-} \ni^F _{-} : GrwT \leftrightarrow Fr}{\forall GwT : GrwT; F : Fr \bullet GwT \ni^F F \Leftrightarrow GwT \ni^{SG} (\bigodot^{SG} F)}$$

relation($_{-} \dashv^F _{-}$)

$$\frac{_{-} \dashv^F _{-} : Fr \leftrightarrow Fr}{\forall m : GrM; F_s, F_t : Fr_0 \bullet F_s \dashv^F F_t \Leftrightarrow fet F_s \neq \emptyset_{GM} \wedge ((\bigodot^{SG} F_s, fet F_s) \dashv^{SG} (\bigodot^{SG} F_t))}$$

relation($_{-} \Vdash^F _{-}$)

$$\frac{_{-} \Vdash^F _{-} : \mathbb{P}((GrwET \times Fr_0) \times (GrwT \times Fr_0))}{\forall GwET : GrwET; F_s, F_t : Fr_0; GwT : GrwT \bullet (GwET, F_s) \Vdash^F (GwT, F_t) \Leftrightarrow GwET^{Gw} \ni^F F_s \\ \wedge GwT \ni^F F_t \wedge \text{domg}(GwET^{ET}) =_p \text{domg}((fet F_s) \circ_G (GwET^T)) \\ \wedge GwET^{ET} \in (GwET^G) \rightarrow_G (GwT^G) \wedge F_s \dashv^F F_t}$$

9 Global Fragment Graphs

section *Fragmenta_GFGr* **parents** *standard_toolkit, Fragmenta_Generics, Fragmenta_Graphs*

$$GFGr == \{ G : Gr \mid \otimes(G \bowtie_{Es} (Es \ G \setminus EsId \ G)) \}$$

function($_{-} \dashrightarrow$)

$$\left| \begin{array}{l} \text{--} \dashrightarrow : GFGr \rightarrow V \leftrightarrow V \\ \hline \forall GFG : GFGr \bullet GFG \dashrightarrow = (GFG \leftrightarrow)^+ \end{array} \right|$$

10 Models

section *Fragmenta_Mdl0* **parents** *standard_toolkit, Fragmenta_Frs, Fragmenta_GFGr*

$$Mdl_0 == \{ GFG : GFGr; fd : V \rightarrow Fr \mid fd \in Ns \ GFG \rightarrow Fr \wedge \boxplus fd \}$$

$$\left| \begin{array}{l} mGFG : Mdl_0 \rightarrow GFGr \\ mFD : Mdl_0 \rightarrow V \rightarrow Fr \\ \hline \forall GFG : GFGr; fd : V \rightarrow Fr \bullet mGFG(GFG, fd) = GFG \\ \forall GFG : GFGr; fd : V \rightarrow Fr \bullet mFD(GFG, fd) = fd \end{array} \right|$$

$$\left| \begin{array}{l} mUFs : Mdl_0 \rightarrow Fr \\ \hline mUFs = \bigcup_F \circ \text{ran} \circ mFD \end{array} \right|$$

$$\left| \begin{array}{l} \text{from} : Mdl_0 \rightarrow V \rightarrow V \\ \hline \forall M : Mdl_0; v : V \bullet \text{from } M \ v = (\mu \text{vf} : (Ns \circ mGFG) M \mid v \in fLNs(mFD \ M \ \text{vf})) \end{array} \right|$$

relation($\uparrow \text{--}$)

$$\begin{array}{|l}
\uparrow - : \mathbb{P} \text{ Mdl}_0 \\
\hline
\forall M : \text{Mdl}_0 \bullet \uparrow M \Leftrightarrow \exists UF : Fr_0 \mid UF = mUFs M \bullet \\
\forall p : (NsP \circ fSG) UF \bullet (from M p, from M (\rightsquigarrow UF p)) \in ((- \dashrightarrow) \circ mGFG) M
\end{array}$$

$$Mdl == \{M : \text{Mdl}_0 \mid (mUFs M) \in TFr \wedge \uparrow M\}$$

$$\begin{array}{|l}
\odot^M : \text{Mdl} \rightarrow Fr \\
\hline
\forall M : \text{Mdl}_0 \bullet \odot^M = \odot \circ mUFs
\end{array}$$

function 1 leftassoc $(- \rightarrow_M -)$
relation $(- \Rightarrow^M -)$

$$\begin{array}{|l}
- \rightarrow_M - : \text{Mdl} \times \text{Mdl} \rightarrow \mathbb{P} GrM \\
- \Rightarrow^M - : \mathbb{P}((\text{Mdl} \times \mathbb{P} GrM) \times \text{Mdl}) \\
\hline
\forall M_s, M_t : \text{Mdl} \bullet M_s \rightarrow_M M_t = \{m : GrM \mid \\
\exists UF_s, UF_t : Fr_0 \mid UF_s = mUFs M_s \wedge UF_t = mUFs M_t \bullet m \in UF_s \rightarrow_F UF_t\} \\
\forall M_s, M_t : \text{Mdl}; ms : \mathbb{P} GrM \bullet (M_s, ms) \Rightarrow^M M_t \Leftrightarrow \bigcup_{GM} ms \in M_s \rightarrow_M M_t
\end{array}$$

relation $(- \sqsupset^M -)$

$$\begin{array}{|l}
- \sqsupset^M - : (\text{Mdl} \times \mathbb{P} GrM) \leftrightarrow \text{Mdl} \\
\hline
\forall M_c, M_a : \text{Mdl}_0; ms : \mathbb{P} GrM \bullet (M_c, ms) \sqsupset^M M_a \\
\Leftrightarrow \exists UF_c, UF_a : Fr_0 \mid UF_c = mUFs M_c \wedge UF_a = mUFs M_a \bullet (UF_c, \bigcup_{GM} ms) \sqsupset^F UF_a
\end{array}$$

relation $(- \ni^M -)$

$$\begin{array}{|l}
- \ni^M - : GrwT \leftrightarrow \text{Mdl} \\
\hline
\forall GwT : GrwT; M : \text{Mdl} \bullet GwT \ni^M M \Leftrightarrow GwT \ni^F mUFs M
\end{array}$$

relation $(- \Vdash^F -)$

$$\begin{array}{|l}
\hline
_ \dashv\vdash^F _ : Mdl \leftrightarrow Mdl \\
\hline
\forall m : GrM; M_s, M_t : Mdl_0 \bullet M_s \dashv\vdash^F M_t \Leftrightarrow mUFs M_s \dashv\vdash^F mUFs M_t
\end{array}$$

relation($_ \dashv\vdash^M _$)

$$\begin{array}{|l}
\hline
_ \dashv\vdash^M _ : \mathbb{P}((GrwET \times Mdl_0) \times (GrwT \times Mdl_0)) \\
\hline
\forall GwET : GrwET; M_s, M_t : Mdl_0; GwT : GrwT \bullet \\
(GwET, M_s) \dashv\vdash^M (GwT, M_t) \Leftrightarrow (GwET, mUFs M_s) \dashv\vdash^F (GwT, mUFs M_t)
\end{array}$$